

**BREAKTHROUGH TEAMS & INNOVATION IN ORBIT:
ENTREPRENEURIAL GROUP INITIATIVES IN ESTABLISHED
ORGANIZATIONS**

by

RICHARD (RICH) BONACCORSI

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SCHOOL OF GRADUATE STUDIES

We hereby approve the dissertation of

Richard Bonaccorsi

candidate for the degree of Doctor of Philosophy.*

Committee Chair

Kalle Lyytinen, Ph.D., Case Western Reserve University

Committee Member

Robert Hisrich, Ph.D., Case Western Reserve University

Committee Member

Yunmei Wang, Ph.D., Case Western Reserve University

Committee Member

Satish Nambisan, Ph.D., Case Western Reserve University

Date of Defense

June 26, 2020

*We also certify that written approval has been obtained
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Breakthrough Innovation & Teams in Orbit:
Entrepreneurial Group Initiatives In Established Organizations

Abstract

by

RICHARD BONACCORSI

How do leaders of established organizations effectively manage intrafirm entrepreneurial groups so that they can be successful? Focusing on drivers of breakthrough groups' performance and radical innovation, this research identifies drivers and proposes a multifaceted model providing a pathway for breakthrough business growth in an organically driven mode. The corporate entrepreneurship (CE) and innovation management literature, while helpful in describing the difficulty associated with leading CE and ambidextrous organizations, are focused on wider structural or behavioral solutions to address ambidexterity challenges and product innovation, omitting the unit of analysis—the group/team—and the drivers that influence groups' breakthrough innovation and performance aims in a contemporary context in which organizations' business models can change in the blink of an eye. Using a QUAN→qual design, the research follows an 'inside-out logic' to account for the position of each introduced construct in the nomological network, which include (1) intragroup mechanisms; (2) team boundary mechanisms (boundary buffering and spanning); (3) environmental mechanisms including senior management communication, incentives,

proactive IT stance; and, (4) contextual influences. The prevalence of a mismatch between conventional incentives vis-a-vis team members' breakthrough pursuits is also investigated. Findings reveal a paradox around top management communication, as well as the mediating roles of team buffering and spanning on the relationship between relational climate and outcome variables, radical innovation, and team performance. Overall, results suggest that just having a top team (i.e., good relational climate) is not enough for breakthrough performance, and could be for naught should the team's boundaries not be managed by leaders as to create buffering (from extra-team interference) and spanning (or access to resources) beyond the team. Environmental mechanisms, and particularly proactive IT capabilities and nuanced senior management communication, also matter when it comes to teams' success. It is hoped that this work, comprising novel constructs and a unique team focus in the context of breakthrough innovation and corporate entrepreneurship, will help drive the two streams of research into new directions.

Keywords: radical innovation; team boundary work; breakthrough teams.

INTRODUCTION

Within large organizations, the exploration of new growth horizons is necessary to sustain and fuel expansion. Indeed, the coronavirus era underscores just how suddenly conventional practices, capabilities, and business models can be disrupted and become irrelevant. Every element of an organization's business model—not just product and customer—can change in the blink of an eye. In the post-pandemic world, the map of surviving competitors will change, regulations will be reformed, and distribution channels, as we know a year ago, may no longer be there (Blank, 2020). Organizational leaders are being asked: How have you empowered (cross-functional) teams to tackle these new radical priorities? What works well, and what holds teams back? Are your core competencies digital-ready, adaptable, and robust? The new normal is that 'incremental improvements are not enough' (du Toit, Debruyne, Fruechtel, & Brookshire, 2020)—all placing a new emphasis on creating cross-functional teams which will work on breakthrough initiatives that help pave the way for new business models, products, and markets.

For decades, the opportunities and challenges facing intrafirm entrepreneurial groups within large, established, incumbent organizations have been studied in the fields of corporate entrepreneurship and innovation management. The need for corporate entrepreneurship (CE) has often been harkened in the news, cautioning that businesses that have been around for a while are 'under siege from an army of local insurgents.' The world's big brands must reorganize and change radically or be doomed (Chaudhuri, 2018).

Entrepreneurial teams within *established* organizations that successfully implement entrepreneurial activities find themselves in opening new horizons that can create a myriad of benefits to the organization, including new streams of revenue and profits, sources of innovation, cultural exchange, and renewal as well as retention of motivated and loyal entrepreneurial talent. Large established firms, after all, have the resources, experience, and talent that would seem to give them an edge. With such strengths, it seems reasonable that established organizations would master the identification and execution of breakthrough initiatives and do everything they can to support the success of their entrepreneurial initiatives. Indeed, studies about entrepreneurship in corporations—also known as corporate entrepreneurship (CE)—show positive links to performance (i.e., Antoncic & Hisrich, 2001), and new products (i.e., Goodale, Kuratko, Hornsby, & Covin, 2011), processes (i.e., Peltola, 2012), markets (i.e., Zahra, 1991), and business models (i.e., Kuratko, Covin, & Hornsby, 2014). By definition, CE can stretch beyond current organizational products and services, technologies, norms, orientations, structures, or operations and push the organization into new directions (Antoncic & Hisrich, 2003).

The focus of this thesis revolves around the research question (RQ) on how to effectively manage teams for breakthroughs. The audience includes not only researchers but also both group and senior managers. As such, while the focus is on drivers of group-level outcomes while devoting attention to the overall organizational context in which the group is embedded.

Managing breakthrough groups within the context of established incumbent firms is much easier said than done. Although most executives would agree that they depend on

innovation for their long-term strategic success, more than 50% report that they have a poor innovation process, and less than 18% believe their own innovation strategy provides a competitive advantage (Koetzier & Alon, 2013; Kuratko et al., 2014). Intrapreneurial projects fail between 70% and 90% of the time (Altringer, 2013). That is, there is no shortage of evidence showing that efforts to stimulate intrapreneurship fall short of expectations (Altringer, 2013). What seems like an unfair advantage for entrepreneurial corporate ventures actually loads the effort with inherited, inflexible liabilities that will block the search for new radical ventures (Blank, 2014). The entrepreneurial groups are forced to go to market in the same way that existing businesses do, which will completely undermine the effort (Macmillan & McGrath, 2003). They easily get absorbed into the company's cross-currents, smothered by bureaucracy, inertia, old practices, politics, and incentives that favor the existing business. Without regard for the distinctive and enduring organizational differences that these initiatives call for and can provide for, the entrepreneurial initiatives will become diluted as they face the powerful forces of 'tradition' (Kotter, 2007).

Sudden jolts to established business models caused by the recent pandemic have caught millions of organizations unable to deal with the rapidly changing business and technology environments. Notwithstanding the pandemic, many established organizations have been trying to align their organizations with novel entrepreneurial traits of small start-ups. A shift from hierarchies to teams has been well underway for some time. In a 2019 survey by Deloitte, 31% of respondents said that they operate mostly or almost entirely in teams. Only 11% of respondents said that their rewards systems were highly aligned with driving organizational goals. Furthermore, while technologies are making

teamwork easier, most organizations have not refreshed leadership, job designs, and rewards that align with the new ways of operating (Deloitte, 2019).

Mirroring the trend, internal, organic corporate entrepreneurship, which has its roots in the idea of transformational change in corporate culture (Kotter, 2007), is on the rise and create powerful dynamics for leaders to manage effectively. Entrepreneurial initiatives take place in highly dynamic and less bureaucratic contexts where colleagues are spread out and work remotely following forces arising from global competition as well as sudden shocks created by pandemics, regulation, or other crises. In this context, increased significance is placed on understanding how intrafirm entrepreneurial groups can be buffered from the interference and how they can span boundaries for resources, skills, and ideas. The gap between desired team behaviors and outcomes and standard, organization-wide incentives ways of operating is wide and growing. Advances in IT have constantly transformed business processes and models. While these solutions close some communication channels with colleagues and senior leaders, they also open others. In short, these radical changes suggest taking a fresh look at what are the antecedents of successful entrepreneurial initiatives in established companies.

Having personally designed and led multiple successful groups in established organizations towards new horizons on a worldwide basis, I was able to see up close that even the best local and global growth initiatives do not anticipate their potential when the internal company climate is not conducive to the groups' success. Unproductive sources of 'help' range from misplaced team members, standard operational and HR 'processes,' and 'good' advice from senior leaders, to inflexible rewards, incentives, and IT constraints, all channeling behaviors to customary ways of doing business. More

specifically, several categories based on my personal experience emerged as critical drivers of the groups' success. These included (1) team dynamics, or internal relationships and change-oriented tendencies occurring within the group boundary (intragroup dynamics); (2) team's external dynamics, or its relationships with other stakeholders within the company (external relationships and resources); (3) degree of inclusion or separation of the groups—which swung back and forth between complete integration with the core business to physical separation, known as skunkworks; (4) use of human resource legacy practices (i.e., incentives, recognition, and rewards) for groups tasked with exploring *non-standard* activities; and (5) IT platforms (legacy platforms usually as inhibitors and new technologies as enablers of customization). Each dimension represents one of the embedded layers of drivers that will be addressed in this inquiry.

As mentioned, the overall research inquiry that follows will be guided by the following research question (RQ): How do managers/leaders of established organizations effectively manage intrafirm entrepreneurial groups so that these groups can be successful? As suggested by the motivation and identified gaps, I aim to identify and validate a multi-level multi-dimensional model of antecedents of intrafirm entrepreneurial team success. Such groups pursue broadly radical innovative initiatives in the context of the established firm. This study focuses on key drivers facing practitioners, as raised by Ashkenas (2011), when they seek to improve the performance of such themes. These include the group's relational and boundary interactions, as well as incentives and information technology support. Many of these have not been addressed in the extant scholarly literature in the context of intrapreneurial terms. In particular, this study takes an inside-out examination of business drivers and begins with intra-group

competencies and proceeds to group boundary interactions, which is followed by extra-group and contextual drivers. This inside-out logic provides the organization for my discussion of the chosen constructs and related hypotheses that follow in this dissertation.

ROADMAP

The organization of the remaining sections of this research is as follows. As a preface to the literature review, I outline the context and boundaries and clarify definitions to be used throughout the dissertation. The subsequent section starts with an overarching theoretical framework based on what we know about dynamic capabilities and organizational ambidexterity as it relates to the RQ. I proceed to hone in on drivers of entrepreneurial group performance by presenting literature from corporate entrepreneurship. Gaps in the theories then pave the way for further literature review. Salient gaps in the management innovation and CE literature include: (1) the group, or team level of analysis; (2) outdated CE antecedents; (3) unsuitable dependent variables. As a result, the literature review continues—outside these domains—identifying and addressing specific influences (or antecedents) of my outcome variables, radical innovation, and performance of groups within organizations. This latter section of theory review imports literature on teams, relational climate, and rewards and incentives, and information technology.

Focusing on drivers of breakthrough groups' performance and radical innovation, the next section proposes a hypothesized multifaceted model of antecedents of group radical innovation and performance and is followed by research design and methods. This section includes examples of the entrepreneurial groups participating in the sample and the use of interclass correlation coefficients to aggregate teams. Supplemental analyses

comprise a post hoc analysis of mediation and an exploration of specific incentives and rewards. In the final section, a practitioner model with recommendations will be summarized, and suggestions for future studies will be proposed.

BOUNDARIES, DEFINITIONS, AND CONTEXT OF INTEREST

In his paper called “Predictors and Financial Outcomes of Corporate Entrepreneurship: An Exploratory Study,” published in 1991 in the *Journal of Business Venturing*, Shaker Zahra triggered a number of subsequent studies by a broad range of scholars who tried to understand which factors influence the financial outcomes associated with corporate entrepreneurship. One contribution of his seminal paper was to provide a definition of corporate entrepreneurship that I will adopt next. He characterized corporate entrepreneurship as follows: “Corporate entrepreneurship may be formal or informal activities aimed at creating new businesses in established companies through product and process innovations and market developments. These activities may take place at the corporate, division (business), functional, or project levels, with the unifying objective of improving a company’s competitive position and financial performance” (Audretsch, 2015: 723). This definition provides a reference point for defining the boundaries of this inquiry across dimensions discussed herein, including (1) the organizational setting; (2) the level and unit of analysis; and (3) type of innovation.

The organizational setting focusses on entrepreneurship within established organizations. With regards to the various levels of where entrepreneurial activities take place (i.e., corporate, division, project levels), I am most interested in a subset of the organization known as the cross-functional group (used here synonymously with *team*), as differentiated from the far more prevalent studies on individual and organizational

levels of entrepreneurship. Therefore, the unit of analysis is the cross-functional group. These are separate dedicated groups, teams, projects, or ad hoc units pursuing non-incremental innovation within established companies. They could be focused on early identification and pursuit of radical new ideas and or be seeking ways to transfer them in later stages towards exploitation and business growth. In this way, a temporal element of the radical initiatives includes those in the early stages of formation and execution, once feasible breakthrough initiatives are identified. Capturing cross-functional groups pursuing radical initiatives (vis-à-vis mature organizations) also allows for the inclusion of several organizational alternative forms reflecting the debureaucratization of organizations. The setting within an established organization illuminates the inevitable presence of inertial forces, contradicting learning modes (Tushman, Smith, Wood, Westerman, & O'Reilly, 2010), and eventual linkages needed to shift to a larger exploitative innovation-driven organization characterized by ambidextrous organizations. This context stands in sharp contrast to studying organizations organized around a uniform 'innovation idea' that ties together the entire organization, whether a start-up tech company or an established organization.

As for types of innovation, this inquiry centers on types of innovations that fit into the context of radical entrepreneurial activities within established organizations. The innovation teams are tasked to... "seek growth options *beyond* conventional new product development that leads to incremental changes in current product lines" (O'Connor & DeMartino, 2006: 475). The type of innovation of interest is exploratory, or radical innovation (RI), and it provides a non-incremental pathway for business growth in organically driven mode through the creation of new lines of business and typically

results in the creation of entirely new markets (O'Connor & DeMartino, 2006: 475). This context contrasts with studies of new product development and associated technology innovation, that are primarily driven by functional departments, like R&D.

Having adopted the definition of radical innovation reflects the heterogeneity of the context of interest. From a scholarly perspective, it overcomes hurdles that many researchers in corporate entrepreneurship (Phan, Wright, Ucbasaran, & Tan, 2009) have observed for not dipping into innovation literature. They have held that radical innovation and corporate entrepreneurship are distinct phenomena (the former refers to innovations from technology push, while the latter usually refers to innovations driven by market opportunities (Phan et al., 2009: 204). As such, this inquiry connects the two domains and draws from some others (i.e., organizational behavior in teams) to address a practitioner-driven problem in contrast to engaging in a singular disciplinary theory-driven study.

Other terms used to describe the kind of innovation that I am interested in are 'discontinuous,' or non-incremental innovation which relies on a firm's capabilities to explore into future opportunities (Levinthal & March, 1993; March, 1991) and which extends or deviates from a firm's existing technical trajectory and/or move it into different markets (Abernathy & Clark, 1985; Eisenhardt & Tabrizi, 1995; Venkatraman & Lee, 2004). O'Connor and McDermott (2004) add that radical, breakthrough, step out, game-changing innovation are all monikers adopted to identify projects and initiatives whose stretch objectives are to create something original to the world offerings and, concomitantly, build up whole new lines of business for companies. We will use interchangeably these terms describing non-incremental innovation.

THEORY REVIEW

In this section, I identify and review three streams of the literature upon which my research draws, builds, and fills gaps.

Dynamic Capabilities

The theoretical roots of dynamic capabilities (DC) are many and include resource-based view, knowledge-based view, behavioral theory, and evolutionary economics, among others. The use of DC in the innovation management and corporate entrepreneurship literature is far-reaching, and no doubt plays an important role in providing overarching insights to this study.

Moving beyond the static resource-based view (RBV), DC generally addresses how firms gain and maintain competitive advantage based on their unique resource configurations. Dynamic capabilities emanated from the view that winners in the global marketplace demonstrate timely responsiveness and rapid and flexible product innovation, along with the management capability to coordinate and redeploy internal and external competences swiftly (Carlsson, 2016). Dynamic capabilities have been cited in relation to studies like this one in which we have learned that radical innovation within the organization is a vehicle for the creation of new knowledge and related capabilities, and these innovations are a product of knowledge flows within and without the organization. The application of dynamic capabilities to the internal organization and innovation is even more relevant. The research, ‘organizing for radical innovation’ notes the importance of simple rules, rather than elaborate processes, and highlights the importance of organizational structures that legitimize transformational experiences (King & Tucci, 2002), situation-specific learning (Eisenhardt & Martin, 2000),

experience accumulation (Argote, 1999) and simple, appropriate rules (Dougherty, 1995) to help build a capability that is not lost from one radical innovation champion to the next (O'Connor & DeMartino, 2006). This latter point suggests specifically that it is important that connections, or bridges, are built between innovative units and core businesses as to allow the exchange and transfer of knowledge and competencies resulting in robust capabilities to draw upon. Furthermore, the role of senior managers in seizing opportunities through the orchestration and integration of both new and existing assets to overcome inertia and path dependencies is at the core of dynamic capabilities (O'Reilly & Tushman, 2008: 187)—a point that resonates with the action-orientation of our research question.

Sorting Out and Focusing Dynamic Capabilities: Organizational Drivetrain

The work by Di Stefano, Peteraf, and Verona (2014), *Organizational Drivetrain*, provides a detailed description and metaphor for the mechanisms that 'bridge' the individual and organizational levels of analysis and of simple and complex rules found in the two settings. Moreover, the framework allows us to position this inquiry within a more comprehensive understanding of DC and to avoid 'cherry-picking' of the numerous citations using DC that have flooded various research streams.

Amid the confusion surrounding the DC construct, Di Stefano et al. (2014) clarify, define, and present DC in a coherent whole. More specifically, the authors highlight the heterogeneity of dynamic capabilities used in the extant literature through an illustration of two seminal yet divergent views. Given the action orientation of our research goal, and the incompleteness of any one theoretical model illuminating our study, it is worthwhile to (1) point out relevant orientations of this research using the five

components; (2) to leverage tenets of DC to build a foundation for our practitioner model which will expand and modify a metaphor of the organizational drive-train; and (3) identify gaps relevant to our research aims that can be filled by other theory and frameworks.

Di Stefano et al. (2014) draw five structural components of dynamic capabilities, summarized in Table 1, to come up with a way to combine the divergent views into a coherent whole. I have shaded the perspective(s) that this study takes, if there is a leaning in one direction guided by our stated research question and context. First, for the nature of the concept, the first component, and in line with the action and practitioner-oriented ‘link to action’ of this research, we lean in the direction of a focus on a “set of specific and identifiable processes.” This contrasts with a view of dynamic capabilities as a latent action, or ability. Second, concerning views about the ‘agent,’ dynamic capabilities vary between the role of the decision-maker versus the organization. While these two views are not incompatible, the managerial view, which emphasizes the decision-maker’s pivotal role, is the one of most interest for this study’s practice-oriented aim to produce a practitioner roadmap for effectively managing breakthrough groups. This view is conferred by Zahra’s view, who defined dynamic capabilities as “the abilities to reconfigure a firm’s resources and routines in the manner envisioned and deemed appropriate by its principal decision-maker(s),” (Zahra, Sapienza, & Davidsson, 2006: 918). Another component, the ‘action,’ refers to whether dynamic capabilities change an existing base or create something new, with some researchers interpreting dynamic capabilities as covering both of these areas. The latter, that DC covers both the existing base and new creations, is the approach this study takes. What matters is that the

innovation is non-incremental, as discussed in the boundaries of this research. An illustration of changing the existing base in my dissertation is the groups' launching core capabilities in new markets. An example of a group's pursuit falling into the category of 'creating something entirely new,' is that of the creation of a media company, when the established business is narrowly focused on the publishing of books.

The authors connect the two views by recognizing the views are connected to a larger interconnected system, which consists of simple rules and complex routines. The interconnectedness, reliance on tacit knowledge, and complexity of a system involving a variety of moving parts suggest that the real source of sustainable competitive advantage for an enterprise is the difficulty of imitating and substituting for the entire dynamic bundle (Di Stefano et al., 2014: 320) that the system represents. In this dynamic system, dynamic capabilities involve both individual and organizational levels of analysis because individuals set the context in the form of simple rules, which then enable and constrain a set of complex routines at the organizational level.

The authors propose an organizational drive-train metaphor. In this metaphor of a bicycle drive-train, the front gear is consists of simple rules which are more likely found in developing something new. The front gear transmits power to the freewheel (complex rules of the organization), which is driven by the rider (top management). The bicycle chain links the two levels of action.

Overall, the contributions of dynamic capabilities, as described by Di Stefano et al. (2014), contributes in multiple ways to this study. Namely, it underscores the importance of connecting different aspects of the organizational system for strategic reasons and discusses the role of simple rules, experience, and learning as a pattern for

adapting radical innovation in a complex system. More specifically, it implies the need to study influences at multiple levels, including senior managers who integrate activities at an organization level. Furthermore, dynamic capabilities, as articulated by the authors, can delve into the micro-level of routines and processes within the organization, in contrast to more abstract and reactive uses of dynamic capabilities as a ‘latent ability,’ for example.

The drawback of the organizational drive-train research and metaphor for the purpose of this research is threefold. First, the group level of analysis is not included in the drive-train metaphor. My proposal, which looks at the relatively unexamined group levels driver of innovative pursuits, rather than individual level, questions the assumption about the dividing line where the simple rules complex routines are carried out. To my mind, and included in our data set, are cases for which it is reasonable to assume that groups creating and combining new and existing capabilities through rules which are both simple and complex. This presents us with the opportunity to develop a model that shows an additional and more equal, less hierarchical relationship between the group and the organization levels. Secondly, the reference to micro-levels of routines does not address the specific kinds of group routines or mechanisms that members and leaders can manage for effectiveness, aspects that I will probe through additional literature streams. Finally, I will discuss and modify the role of top management as the rider. My modifications respond to the drive-train’s authors ‘hope that others might find our efforts thought-provoking enough to merit further development’ (Di Stefano et al., 2014: 318).

Table 1: Defining Dynamic Capabilities Using Differing Views

Component of DC	One view	'Opposing' view
Nature of the construct	Ability (latent)	Process/routine for effectiveness
The Agent	Pivotal role of Managers	Organizations/Firms
The Action	Change existing base	Create something new
Object of the Action	Competencies	Opportunities
Aim/Purpose	Adapt to changing conditions; compete over time, AND...	Achieve competitive advantage/ increase effectiveness;

Shaded cells indicate the orientation of this research

Source: Di Stefano et al. (2014)

Zahra (2015) applies a subset of dynamic capabilities (DC) to corporate entrepreneurship, which is particularly relevant to study. As discussed, I am interested in where and how groups carry out CE initiatives. Zahra's subject of study is similar to my research interests in a couple of ways. He uses a heterogeneous definition of CE initiatives, which include a wide variety of non-incremental initiatives. Secondly, his focus unit of analysis lies in the space between the organizational and individual levels, the extremes represented in the extant literature. The author's examination of 'entrepreneurial hubs' is the closest to my focus on entrepreneurial groups found in the CE literature, as far as I can tell. He looks beyond formal organizational structure, "where different coalitions of interested people" take part in the entrepreneurial process."

How can we apply lessons to this thesis? First, his work specifically addresses the valuable knowledge that is generated in CE initiatives. He considers the knowledge "the foundation of competitive superiority," which, if managed, builds linkages within and across the different units of an organization. By adopting a subset of DC that puts learning and interactions at the center, Zahra's frameworks (Zahra, 2015; Zahra,

Sapienza, & Davidsson, 2006), helps adapt my group based radical innovation model that focuses on the implementation of corporate entrepreneurship. The creation of knowledge is pervasive across levels and organizational boundaries, which not only confirms the needed linkages within the organization described in the earlier discussion of DC literature but also expands on it by addressing specific mechanisms (knowledge sharing and exchanges; interactions) and organizational layers (multiple) to derive and share knowledge. In short, at the core of knowledge creation and knowledge conversion are interactions among different players, which result in refined ideas, which then get passed on (as new knowledge) to the next phase of the process. Through interactions, participants absorb, refine, validate, and produce knowledge that informs innovation decisions. In this way, his work identifies the potential that interaction and relational aspects within and beyond the group, and identifies important sources of knowledge, often not tapped by senior and middle managers who are unaware of ‘knowledge’ creation that happens below the formal organizational-level radar.

Ambidexterity – Context and Clashing Learning Logics

The context and problem of practice can be viewed under the lens of differing learning logics. Above, I briefly noted the presence of competing and paradoxical organizational learning logics requiring firms to improve what the organizations are doing on the one hand, and to create radically new and markets and products on the other hand, pose daunting challenges to leaders. These challenges have been discussed widely in the ambidexterity research originating mainly from technology management, strategy, and innovation literature. These studies probe alternative structural and temporal forms of ambidexterity to organize radical innovation initiatives as they are launched, cultivated,

and maintained within a firm while it still exploits its ‘old business.’ Exploitation is driven by a regime of continuous, incremental change anchored on a given technical/customer trajectory. In contrast, exploration is a learning mode driven by variability (Tushman et al., 2010: 1358). The paradoxes associated with opposing businesses within an incumbent organization are described as counterproductive and potentially destructive as competency and speculation traps. Given the paradoxical nature of ambidexterity, exploitation will also drive out exploration and vice versa (March, 1991; Tushman & O’Reilly, 1996). O’Connor and McDermott’s (2004) studied radical innovation in established companies, and they summarize this problem as the presence of “numerous antibody mechanisms are embedded in the organization that constantly block and thwart the advancement of fundamentally new ideas, and sometimes, view maverick individuals as too painful to tolerate.” This growing body of research on ambidexterity has significantly contributed to describing the context in which radical intrafirm innovation teams operate and identifying contextual and structural challenges associated with the study’s problem of practice

Studies contrasting structural and contextual ambidexterity further suggest clashing modes of learning and transferring knowledge, with implications on the role of group and senior-level managers. Exploitation involves learning from a top-down process in which the top management moves to institutionalize the routines and behaviors that are most suited for refining current competencies (de Bruijn, 2011: 22). In contrast, exploration generally involves bottom-up learning in which the top management team is persuaded to change old routines and commitment to new action (Floyd & Lane, 2000; Lubatkin, Simsek, Ling, & Veiga, 2006; Rivkin & Siggelkow, 2003). Mom, van den

Bosch, and Volberda (2007) found empirical evidence that bottom-up and horizontal knowledge inflows are positively related to a manager's exploration activities and create a variety of experience, which is associated with exploration. Their empirical evidence also showed, by contrast, that top-down knowledge, inflows associated with knowledge coming from persons or units at higher hierarchical levels than the manager, were associated with exploitation.

This distinction in learning suggests a stark contrast in roles of group and senior managers in effectively managing groups, noting the importance of differential learning logics. It highlights the need to match the right knowledge flow—one based on bottom-up and horizontal flows—to exploratory groups' activities and top-down knowledge flows for exploitation activities.

Ambidexterity – A Focus on Antecedents, Outcomes, and Unit of Analysis

The number of studies in leading management journals that explicitly refer to the ambidexterity concept has increased from less than 10 in 2004 to more than 80 today (Raisch et al., 2017: 685). This subset of studies sheds light on certain, specific antecedents, or factors, influencing outcome variables. Examples of these research papers are listed in group A.

Structural ambidexterity. Structural and contextual ambidexterity represent the two widely accepted meanings of organizational ambidexterity (Gibson & Birkinshaw, 2004). The one that exists in the firm's structure refers to as the structural ambidexterity. Structurally ambidextrous firm separates the explorative from the exploitative unit, each with different management, processes, structures, and cultures (O'Reilly & Tushman, 2004). Previous findings suggest that applying cross-functional structure (for

ambidexterity) to radical NPD process, the effect on breakthrough innovation performance is positive (Zaidi, Faizal, & Othman, 2015).

Table 2: Group A: Ambidexterity - Focus on Singular Antecedents, Mixed DVs, Unit of Analysis

Article	Unit of Analysis	Context/Sample	Role of Exploration/ Exploitation	DV	IVs of Interest	Duality v. Specialization
Hill and Birkinshaw (2014)	Venture units	Corporate Venture Units	Structural	Ambidexterity/ Survival	Sr. Management relationships (Relational)	Duality
Gibson and Birkinshaw (2004)	Business units	41 business units of 10 corporations	Exploration and Exploitation as mediators	Performance	Social Context (support, discipline, stretch, trust) Performance mgt	Duality
Tushman et al. (2010)	Business units and entire single product organizations	Product innovations as the focus	Both – emphasize difficulty of exploration	Innovation outcomes	Structural – 4 org designs; Sr. mgmt. : Innovation streams by type (architecturally, discontinuously, incrementally)	Duality
O'Reilly and Tushman (2008)	Organizations or BU still ambiguous	Whether it is org or BU level, or sequential or in parallel is ambiguous.	Ambidexterity as a dynamic capability which falls on leadership	Ambidexterity (ability to do both means organizations can adapt	Leadership (actions, design processes, choices that matters, not structure or design.	Duality
O'Connor and Demartino (2006)	Org	12 MNCs, new and established R&D	Exploration	Radical Innovation capabilities	Org structures – 7 models;	Radical innovation
O'Connor and McDermott (2004)	RI Projects in R&D functions	10 established US based firms	Exploratory projects	Radical Innovation	Incentives Rewards, team composition and formation	Radical Innovation
Glaser, Fourné, and Elfring (2015)	Org/BUs	34 BUs within single logistics firm	Exploratory innovation	Exploratory Innovation RI Jansen	TMT v MM Boundary spanning	Exploratory
Zaidi et al. (2015)	Product and PMs of New Products	New Product Development; Manufacturers	As Independent variables	Financial and Non-Financial Performance	Structural and contextual ambidexterity	Comparison of structural and contextual
Bruijn (2011)	Business unit	Financial Services BU of large group managers/non-managers	Contextual	Contextual ambidexterity/ Exploitation and exploration combined.	Social context; knowledge inflows	Duality
Wang and Rafiq (2014)	New Product innovation in Business Unit	Business Unit	Contextual, ambidextrous organizational culture	New product innovation (Radical, incremental innovation, speed to market	Contextual ambidexterity; ambidextrous org culture.	Duality

Remedies from much of this extant scholarship furthermore often suggest structural separateness (i.e., Glaser et al., 2015; Hill & Birkinshaw, 2014). Herein lies a second gap as it relates to my study. Reorganizing and physically removing groups from the core fall short in a number of ways. First, they fail to account for the intra- and extra-group dynamics and relational aspects that promote group success within and beyond the (exploratory) groups. Second, they lack the nuanced extent of separateness from and connectedness to the established core organization. As helpful as the extant literature is with regard to formal structural organizational forms and their impact on innovation, this thesis examines exploratory groups of various forms as long as they are cross-functional groups pursuing non-incremental innovation. It is in these settings, which includes traditional ‘skunkworks’ as well as early and late stage, formal and informal groups, where groups of professionals pursue non-incremental innovation initiatives. In this regard, our view is aligned with O’Reilly and Tushman’s (2008) findings that ambidexterity is not ‘simply a matter of organizational structure and design (p. 201), or that that exploratory sub-units need to be spun out (Gupta, Smith, & Shalley, 2006). Instead, we shift the emphasis to a ‘leadership task.’ Notwithstanding our agreement with this view on this topic, the literature does not say enough about what that means at the level of the exploratory unit, and what managers and leaders do to effectively manage and lead exploration. After all, many of the corporate entrepreneurship activities will traverse the informal and formal organizational structure where different coalitions of interested groups take part in the process at different times and for different reasons (Zahra, 2015). They are not necessarily separated by formal boundaries, or ‘walls.’

Contextual ambidexterity. It is not sufficient to address initiatives that are implemented in ‘stand-alone’ structures alone. A second meaning of ambidexterity, contextual ambidexterity is helpful in this regard. Defined as ambidexterity that lies in the behavioral orientation is referred to as contextual ambidexterity (Luzon & Pasola, 2011) and is helpful in that it provides social and ‘soft’ mechanisms in contrast to systematic and formal mechanisms discussed so far. This type of ambidexterity mirrors the practitioner-oriented literature that underscores the dangers of overemphasizing formal organizations as studied in structural ambidexterity scholarship, and calls attention to more complex webs of social relationships. Ashkenas’s (2011) article, *Reorganizing? Think Again*, and Leodolter’s (2017) article *Will Organizations Emerge as “Hybrid Intelligences” from the Digital Transformation?*, point out the problem that functional organization is only partly determined by structure. The web of relationships, incentives, sharing of resources, and cross-functional interactions, lack of support for information systems, overemphasis on building consensus may matter more. An overemphasis on structure is further problematic in an era of porous organizational boundaries, ever-moving loci of entrepreneurship, and of surges in ‘virtual distancing,’ in which proximity of colleagues and physical separation of boundaries becomes increasingly irrelevant.

Previous studies have demonstrated the influence of cross-functional contextual ambidexterity on firm performance (Gibson & Birkinshaw, 2004). Contextual ambidexterity is a meta-level capacity where the simultaneous pursuing of explorative and exploitative activities within a single business unit is built on the processes or systems that encourage individuals to divide their time between activities. Individuals split their time between alignment focused (exploitation) and adaptability focused

(exploration) activities. Simply stated, it is the “simultaneous exploration and exploitation within a business unit” (Wang & Rafiq, 2014: 59).

Applied to the context of this thesis, this describes a scenario in which radical innovation groups studied may not be dedicated to the exploratory pursuit, and instead may be splitting their time with other (exploitation) activities. Consider the cross-functional team at a publishing company that is committed to digitizing all its content and transforming its business into a media company. The cross-functional group (included in our sample) meets regularly as it launches a business model that will include the creation of all functional areas, positioning the company in an entirely different industry, the digital media industry. The individuals exhibit contextual ambidextrous because the members also simultaneously spend time on the ‘old’ business—producing and marketing printed books in the traditional medium as paperbacks or hardbound. That is, the members experience contextual ambidexterity at the individual level. The CEO, in this example, oversees contextual ambidexterity as manifested at the organizational level, as the person who is responsible for managing the linkages of both businesses. This is why the literature says that contextual ambidexterity is needed to create integration between dual structures with behavioral and social means (Andriopoulos & Lewis, 2009). For, without contextual ambidexterity, structural ambidexterity could lead to isolation, although the combination of them as complementary has been established (Zaidi et al., 2015).

Take-aways from structural and contextual ambidexterity. The cross-functional groups of this study sometimes are structurally separate or not, making both contextual and structural ambidexterity relevant in describing the context. These streams, as

discussed, further contribute to our RQ of how to effectively manage breakthrough teams and initiatives by underscoring the perspective of senior leaders' role in integrating dual structures from the organizational level perspective. Jansen, Tempelaar, van den Bosch, and Volberda's (2009) study, more specifically, examined specific types of integration mechanisms and showed the positive influence of senior team social integration and cross-functional interfaces are both related to achieving ambidexterity.

Overall, the rich ambidexterity research is helpful in addressing, from a top management perspective, the not-so-well understood management and integration of various subsystems of the organization and is helpful in developing a model for both group and senior management, a secondary goal of this thesis.

However, largely missing from ambidexterity literature is the study of specific mechanisms that organizational leaders could tap to influence performance outcomes of the exploratory teams. The popularity of the achievement of ambidexterity, as evidenced by its use as the dependent variable is different from the entrepreneurial group's aim, in the context of this thesis. For example, in the case of studies on contextual ambidexterity, the outcome variable includes items such as: "seek out opportunities to combine your efforts with others," and 'act as a multi-tasker who is comfortable wearing more than one 'hat.' Perhaps obviously, these are not indications of groups' exploratory success or aim according to the context and problem of practice described.

Secondly, the product focus vis-à-vis more general innovation or entrepreneurship cannot be underestimated. For example, in Zaidi et al. analysis of structural and contextual ambidexterity, the skills of employees in structural ambidexterity are described as 'focused,' while those contextual ambidexterity are 'more generalists' by

Gibson and Birkinshaw (2004). This does not square with radical innovation groups, as illustrated through examples of the creation of radically different business models. The skills required of a cross-functional group of credit union colleagues, setting up the first, digitized credit union in which all functional areas would be redesigned are notably general in scope. These group members are in roles requiring generalists, or ‘wearing of many hats’ inherent in building a new business model, rather than performing defined roles.

Third, the importance of senior management’s role in social integration, cross-functional interfaces, and balance of exploration and exploitation activities is acknowledged, as evidenced by the finding that innovation streams “actively managed by the senior team were relatively more successful than innovation streams managed by either below or above the senior team.” However, the authors recognize the limitation that senior team integration was not sufficient to host innovation streams’. Advice for senior managers is limited to general guidance such as ‘building into the organizations the internal contradictions necessary to simultaneously explore and exploit’ (Tushman et al., 2010: 1359) without really focusing on leadership and change behaviors to adopt within and beyond levels the exploratory sub-units.

Building a Model - The foundation

The evolution of our model is included in Figure 1. Our metaphor, with simple illustrations, begins by placing two gear wheels side by side (1a). One wheel represents the entrepreneurial group, and the other gear represents the organizational core. With their gears meshed (bumping up against one another), the wheels move in opposite directions. These directions of the wheels represent the friction and contrary ways of the

two wheels. Further, which wheel drives and which is driven can change, as evidenced by groups that later may become the core, or ‘future,’ of the organization, as illustrated by our examples of digital business models in the making.

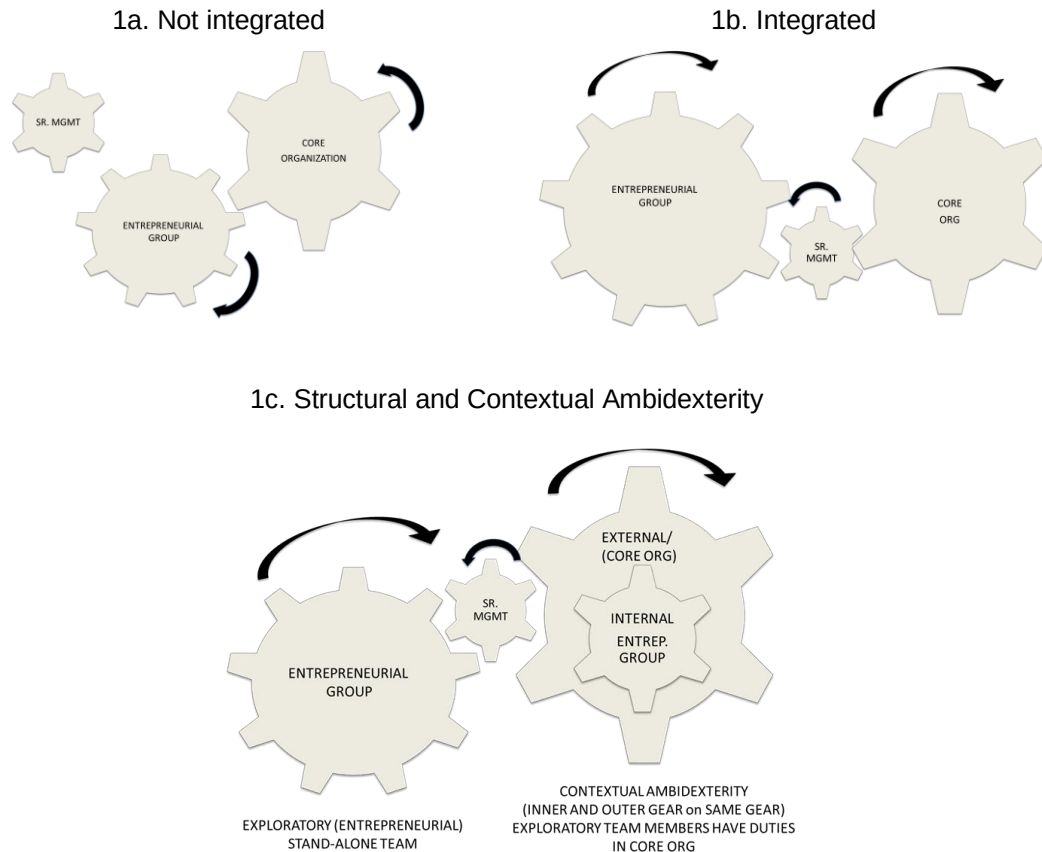
Our metaphor introduces the concept of an idler gear in 1b. An idler gear is a ‘gear wheel placed between two other gears to transmit motion from one to the other.’ It does not alter the speed of the output, but it does reverse the direction it turns.” The idler represents the managing mechanism (i.e., manager, top management) who can capture, manage, and integrate knowledge and relationships of both entities at the organizational level. Altering the direction of rotation of one wheel can put the two wheels in the same direction, illustrating the effect of leadership’s integration. Alternatively, the leader could allow the two wheels to orbit in different directions, or permit the entrepreneurial group to be the drive wheel, most likely after reaching a mature state, as could very well be envisioned by the leaders of the two notable groups building business models of the future for their organizations.

Unlike the bicycle drive-train, our ‘wheels in orbit’ have no one ‘driver’ from above. Instead, the leadership in our model is symbolically placed side by side the group and other organizations, adapting and adopting top-down, lateral, and bottom-up knowledge flows, in proportion to the goals of the subsets—groups, units, core business—within the organization. Of course, various configurations of wheels are possible as ambidexterity can be manifested in multiple ways at one time, orbiting in different directions freely or connected to other subsets of organizational units and teams. The inner/outer gear, 1c, serves as a metaphor for contextual ambidexterity, illustrating

exploratory group members that come together from other exploitative roles to create breakthrough innovation.

This metaphor, that will form the basis of a model for the secondary goal of this research, will be developed and described throughout this literature review and research.

Figure 1: Wheels in Orbit – Part 1



Corporate Entrepreneurship – Multi-Dimensional Models

Like the managerial innovation literature, the corporate entrepreneurship research, outlined in group B, yields rich descriptions of contextually embedded intrafirm innovation teams. Researchers such as Antoncic and Hisrich (2003) and Yariv and Galit (2017) have identified the friction and contradictory aims of established operations vis-à-

vis entrepreneurial groups/ventures within the established organization. This body of work offers definitions that reflect bold and heterogeneous aims of the entrepreneurial groups. Corporate entrepreneurship, also referred to as corporate venturing, or intrapreneurship, was initiated in established organizations for purposes of profitability (Zahra, 1991), strategic renewal (Guth & Ginsberg, 1990), fostering innovativeness (Baden-Fuller & Stopford, 1994), gaining knowledge for future revenue streams (Mcgrath, Venkataraman, & MacMillan, 1994), and international success (Birkinshaw, 1997)—many of which align with the notion of exploration. Corporate entrepreneurship centers on re-energizing and enhancing the ability of a firm to acquire innovative skills and capabilities and deploy them for survival and market growth (Hornsby, Kuratko, & Zahra, 2002).

Vesper (1984) developed early on three characteristics common to corporate entrepreneurship: (1) new strategic direction; (2) initiative from below; and (3) autonomous business creation. The definition that seems to capture the scope of this study in terms of corporate entrepreneurship has been best expressed by Damanpour (1991), who states that “innovation can be a new product or service, an administrative system, or a new plan or program pertaining to organizational members” (p. 556). These works all help define the context and goals of the effort, just as the ambidexterity literature stream did in slightly different words that focus on learning and response variety.

Table 3: Group B: Corporate Entrepreneurship: Multi-dimensional Models of Intrapreneurship

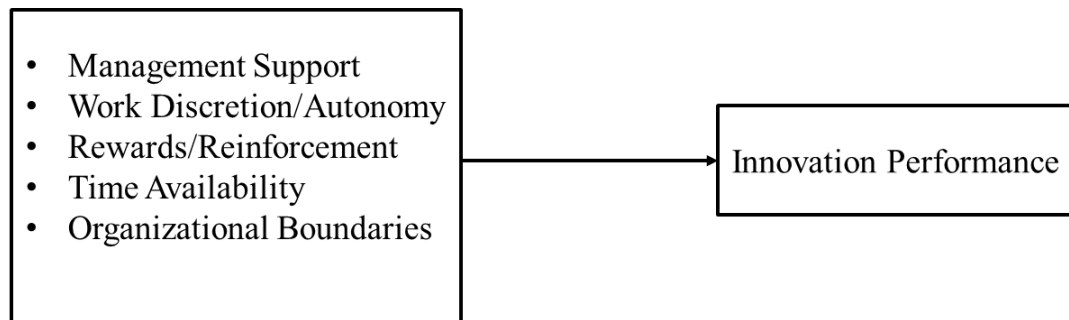
Article	Level of Analysis	Context/ Sample	DV	IVs of Interest	Gaps for this study
Kuratko et al. (1990)	Individual	Diverse Organizations (independent firms) and Business Units	Innovation performance (individuals)	Corporate Entrepreneurship Assessment Instrument (Rewards, upper mgt support, org boundaries, time availability); A systems perspective of drivers.	Not updated; diluted sample - not subsystem of organization (not ambidexterity context).
Hornsby, Kuratko and Zahra (2002) Goodale et al. (2011) Hornsby, Kuratko and Montagno (1999); Hornsby et al. (2002); Goodale et al. (2011)	Individual	Diverse Organizations (independent firms) and Business Units	Innovation performance (individuals), which is product innovation centric.	Refined Corporate Entrepreneurship Assessment Instrument: Top management support, work discretion/autonomy, rewards/reinforcement, time availability, and organizational boundaries; Performance-based rewards construct	Diluted sample - not subsystem of organization (not ambidexterity context). Constructs not updated; Used many years without update; Conflicting findings for performance-based incentives; and only 2 of 5 significant in most recent study. No leadership/change capability at team level

Ireland, Covin, & Kuratko (2009)	Organizational level;	Top level driven CE strategy; CE as strategy	Competitive capability; strategic repositioning	Org vision, architecture	Org wide phenomena instead of our focus - group driven, sanctioned (org level) and unsanctioned initiatives.
Zahra (2015)	informal “hubs”, a unit of analysis that’s akin to ours’ vis a vis individual and org levels.	International and national entrepreneurial hubs	CE member interactions as knowledge and strategy;	Relationships/Interactions, moderately referencing intragroup dynamics (‘relational climate’) and sr. management interactions.	Interactions doesn’t capture leadership or group dynamics and hubs not same as ‘teams’; however, some similarity.

The reviewed materials in this category are helpful in addressing the multifaceted challenges facing managers to successfully manage such radical innovation pursuits. My inquiry's problem of practice—effectively managing entrepreneurial groups—matches the CE literature scope in that much of it centers on the internal organizational environment and antecedents of entrepreneurship in the context established, 'ambidextrous' organizations. In this way, this stream of CE literature most closely mirrors studies in the past concerned with the internal organizational factors that encourage corporate entrepreneurship (Covin & Slevin, 1991).

The work by Kuratko et al. (1990) offers multiple, intra-organizational antecedents that influence innovation. Building on previous work of organizational antecedents of individuals' entrepreneurial behaviors, Hornsby et al. developed a Corporate Entrepreneurship Assessment Instrument (CEAI)—an instrument designed to tap into five key organizational antecedents to corporate entrepreneurship (Goodale et al., 2011; Hornsby et al., 1999, 2002). The antecedents form 'an underlying set of five stable organizational factors that should be recognized in promoting entrepreneurial activities within an organization' (Hornsby et al., 2002: 269) and comprise: management support, work discretion/autonomy, rewards/reinforcement, time availability, and organizational boundaries, as depicted in Figure 2. The model has been used to assess a firm's internal entrepreneurial environment, and its use has shaped corporate entrepreneurship ever since.

Figure 2: A Model of Intra Organizational Drivers for the Corporate Entrepreneurship



Overall, the work underscores the influence of a multi-level, multifaceted approach to CE performance. The influence of top management, for example, is evidenced through a construct used for management support. Defined as ‘the extent to which one perceives that top managers support, facilitate, and promote entrepreneurial behavior; including the championing of innovative ideas and providing the resources people require to take entrepreneurial actions,’ it has been found to have a positive relationship with an organization’s entrepreneurial outcomes (e.g., Antoncic & Hisrich, 2001; Hornsby et al., 2002; Kuratko, Ireland, & Hornsby, 2001; Lyon, Lumpkin, & Dess, 2000; Morris, Kuratko, & Covin, 2008). Hornsby et al. (2002) found there were significant differences on all five factors influencing on entrepreneurial behavior (p. 269) when comparing between the upper-middle level and the middle to lower-middle manager level, which we believe contributes to the complexity and necessity of managing multiple layers of influences on CE initiatives.

The corporate entrepreneurship literature also contains gaps. First, a lot has happened in the corporations since the development of the CEAI in the 1990s and early 2000s. This model leaves out recent phenomena relating to contemporary organizing and

the increasing role of technology. Specifically, boundary buffering and spanning in the context where structural boundaries are increasingly porous, and team members are geographically dispersed, is not recognized. Current challenges and opportunities have also changed due to all-encompassing digitalization. Within larger established organizations, a growing concern is that established companies are increasingly unable to deal with their rapidly changing business and technological environments. New organizational forms are being tried out to emulate entrepreneurial spirit and corps d'esprit of small start-ups. Instead of traditional organizational forms and reporting relationships, more significance can be derived from understanding how entrepreneurial groups can be buffered from bureaucratic interference and how they can effectively span boundaries for new resources.

Secondly, with regards to specific constructs, although the body of work produced an instrument of productive five critical organizational dimensions of CE, the performance-based 'rewards/incentives' factor has resulted in mixed and insignificant results (i.e., Goodale et al., 2011; Hornsby et al., 2002). Moreover, while management support has provided useful contributions over time to CE research, this factor is unclear in specifying which upper management the construct referred and, more importantly, doesn't provide specific actions for upper management to take. As a result of the shortcomings, we have selected a different measure, which will be discussed in the measures section of this thesis. Finally, the dependent variable, "innovation," as commonly used by the aforementioned authors, perpetuates a product focus, not unlike the innovation literature, and is, therefore, ill-suited for our research. This measure consists of items, such as 1) number of new products or services developed, (2) number

of new products or services brought to market, (3) speed with which new products or services are developed, (4) speed with which new products or services are brought to market...” (Goodale et al., 2011). This product-centered measure, which lacks the more general scope of ‘entrepreneurship,’ clearly does not apply to the kind of RI of my dissertation, as outlined in Table C, which by design extends beyond product innovation. Additionally, like the ambidexterity stream, it leaves out a measure for the leadership and change capability at the team level.

A third gap in the entrepreneurship literature relates to its a blind spot in the group/team unit of analysis. Antoncic and Hisrich (2003) already pointed out that most characteristics of intrapreneurship are defined from the standpoint of the whole organization while overlooking supplementary views. Studies that analyze CE related phenomena at the organizational-level include Covin and Slevin (1991), Lumpkin and Dess (1996), and Ireland et al. (2009), and are also reflected in company-wide CE strategies discussed by these authors. These all differ from the study’s level of analysis, which is cross-functional teams within complex organizational settings. Other studies, including those by Hornsby and Kuratko focus on *individual* entrepreneurial behavior. Variations on the individual level include Yariv and Galit (2017) study, which took the viewpoint of employees, rather than that of managers in addressing organizational support for corporate entrepreneurship. While Yariv and Galit’s (2017) adds the lens of the employee, there have no studies that I am aware of that have taken the view of an entrepreneurial group/team, as this study does.

Ireland, Reutzel, and Webb’s (2005) review of entrepreneurship research in the *Academy of Management Journal (AMJ)* also reveals the scarcity of articles that focus on

the group (or team). In the 50 articles reviewed in the period from 1963 to 2005, individual and firm-level analysis dominated the entrepreneurship research. The use of the individual entrepreneur as a level of analysis has been relatively steady. Yet, there has been a small increase in the number of articles (26 of 34) using the firm level of analysis since 1994, supporting the dominance for firm-level analysis. Most shockingly, only *two* studies have used the ‘group’ as the level of analysis—a surprising finding given the origins of corporate entrepreneurship are often a subset of the organization—a team.

This gap in the entrepreneurship literature leads to studies in which survey sample respondents of entrepreneurial ‘groups’ are not properly delineated from respondents of other non-entrepreneurial, organization level respondents who are *not* working on entrepreneurial initiatives. I refer to this drawback as “diluted sample” in the last column “gaps of this study.” By diluted sample, we mean that managers from both “organizations” and business units (not necessarily exploratory sub-units within the organization) were included. Even the seminal research by Goodale et al. (2011), included managers from both “single, independent firm or a business unit within a larger corporation,” leaving it unclear as to the extent to which firms/respondents referred to exploratory/radical initiatives in their responses.

Singular Antecedents of Interest

In describing the multi-theoretic approach to CE research, Ireland et al. (2009) provide a justification such approach “given the diversity of constructs, levels of analysis, and relationships explored in the proposed model, it is doubtful that a single theory or small set of theories can be offered...” (p. 37). Likewise, I adopt focused research from

various disciplines sheds light on specific critical antecedents, even if the unit of analyses is different. These are included in group C.

Table 4: Group C: Literature from Multiple Domains with Substantive Focus on Singular Antecedents of Interest

Article	Unit of Analysis	Context/Sample	DV	IVs of Interest
O'Connor and McDermott (2004)	Established R&D teams	R&D products and processes	Radical Product and Process innovation	Compensation/incentives
Rosen et al. (2013)	Individual	Full and p/t Employees	Scale development content/frequency of idiosyncratic-deals	Compensation/incentives
Faraj and Yan (2009)	Team	Application software teams	Team performance	Boundary Buffering and Spanning
Rochford and Kendall (2016)	Individual	R&D and IT	Product innovation success	Relational Climate (shared vision, mood, energy, compassion)
Rochford and Boyatzis (2016)	Individual, Organization, Team	Conceptual; Workers in orgs with >10 employees	N/A	Relational Climate (shared vision, mood, energy, compassion)
Lu and Ramamurthy (2018)	Organization	IS executives of med-sized firms	Organizational Agility	Proactive IT Stance

Intentional Change Theory (ICT) and Relational Climate

Literature from organizational behavior (Rochford & Boyatzis, 2016) addresses group dynamics and contributes to my discussion on theoretical framing and hypothesis development of how intrapreneurial teams can become more effective.

The theory helps to get down to micro-level analysis of interactions among people who transfer knowledge in innovation teams. ICT places people and *type*, or quality, of interactions at the center of change and learning in team settings. It addresses the presence and importance of empathy, support, and relational energy among members. Additionally, through interactions that take place, it explains the multi-levelness of the model, thereby more specifically addressing the contagion that can be triggered (by senior leaders) to ignite and spread the transfer of knowledge.

Specifically, ICT address the behaviors that cultivate and sustain individual, group, and organizational change (Akrivou, Boyatzis, & McLeod, 2006; Boyatzis & Smith, 2006; Van Oosten, 2006). Boyatzis' writing in *Coaching Teams Through ICT* (2018) underscores the complex system of relationships in which a team is embedded—comprising individuals within the team and dyads, to other teams, organizations, and communities. Each level has people who are leaders and influence the focal team. He posits that ‘If these leaders are resonant, the team has a supportive external environment within which to explore continuing change and learning. If the leaders are dissonant, change will likely be stopped or at least inhibited. This results in a defensive, “That’s not how we do it here” or the “not invented here” claims used to deny the relevance of possibilities. These multiple levels of leadership will dramatically affect the development

of the team and the sustainability of any learning, innovation, or change experienced (Boyatzis, 2018).

Does ICT apply to the group level and across levels? According to Boyatzis (Boyatzis, 2008), desired change occurs at multiple levels of human and social organization. These levels are individual, dyad, or couple (including boss, subordinate relationships); team, group, or family; organization; community; country/culture; and global. Complexity theory and ICT propose that sustainable change at any level must involve developmental work at the levels below and above it (Chen, Kirkman, Kanfer, Allen, & Rosen, 2007; Simsel, Viega, Lubatkin, & Dino, 2005). Effective team development requires effort at the individual level, among various dyads, and at the organization and community levels. According to Boyatzis, it takes effective leadership on many levels to effect sustainable change. In order for this to occur, there must be an agent that carries the “contagion” of emotional and other messages back and forth among the levels. For example, this happens through a team ‘captain’ who keeps people working together, and is the link between the individual and the team change. Meanwhile, the ‘coach’ is the person who links the team with management (Boyatzis, 2008). Resonant leadership accomplishes this between individuals, dyads, teams, organizations, and communities (Boyatzis, 2006). What will I specifically leverage from ICT? Returning to the outer layers of our wheel, leadership capabilities for team behaviors relate to multiple levels of change. Leadership supporting change capabilities happen at the group level, and beyond, from the team to the environmental levels. Likewise, an unsupportive external environment can stop the change capability of the group. Focusing on the group unit itself, the basis for change capabilities within the team, and can be measured by the

team's relational climate by Rochford and Boyatzis (2016), which we expect will influence our outcome variables. Examining the relational climate measure as a basis for a leadership and change capability seems to fill a gap left by the ambidexterity and CE literature streams.

Finally, the word *intentional* of ICT plays an important role in my model, which argues for a continuous, proactive emphasis on corporate entrepreneurship by firm leaders, who intentionally drive change and innovation continuously. This may be somewhat similar to the proactive perspectives of DC, for example, the 'decision-maker as agent' of dynamic capabilities (Di Stefano et al., 2014).

Boundary Mechanisms – Spanning and Buffering

Faraj and Yan's (2009) research on boundary work focuses on software application teams and examines boundary buffering and spanning within organizations of a different setting than my focus. Yet, it contributes in a major way to this study's aims and findings. The extent that the group unit has doors open (for boundary spanning) to the core organization and closed (for boundary buffering or protection) from the exploitative organization seems to be a key to the entrepreneurial group's success. However, measures for these two functions are absent from the ambidexterity literature. This study adopts boundary buffering and boundary spanning (Faraj & Yan, 2009: 604) for this purpose. It is defined as "the extent to which the team encourages its members to solicit critical information and resources, influence important external agents or actors, and build relational networks" beyond the group. Through boundary buffering, mechanisms for achieving proper 'room,' or separateness from the established organizations are captured. As discussed by Faraj and Yan (2009) teams now must

operate in organizations that are becoming “debureaucratized” (Denison, Hart, & Kahn, 1996), boundaryless (Kerr & Ulrich, 1995), network-based (Majchrzak, Jarvenpaa, & Hollingshead, 2007), temporarily structured (Bechky, 2006), geographically dispersed, electronically mediated (Gibson & Gibbs, 2006; Kirkman & Mathieu, 2005), and embedded in a pervasive digital ecosystem (Yoo, Boland, Lyytinen, & Majchrzak, 2012). This profound change away from formal structures and bureaucratic organizing principles raises important questions as to how teams manage boundaries in an organizational environment wherein cross-functional teams are now prevalent (Cappelli, 1999; Deloitte, 2019).

Rewards and Incentives

As discussed above, compensation and rewards has been included in several quantitative studies covering antecedents of CE, but with mixed results and it comes with specific methodological challenges. As discussed above, the CE literature leaves open the question as to whether performance-based rewards drive radical innovation performance. I will settle that score on this point through this research with my empirical analysis. If results show that—yes, they matter—then it raises further questions about practical implications for managers and leaders.¹ So, what is the specific content, or incentive and/or compensation mechanism or tool, which makes a difference for the individuals in such settings and, therefore, makes the groups’ radical innovation initiatives more successful? There is no single answer to this question in the empirical literature we reviewed. What is often missing in these studies and needed is a more refined

¹ Should it matter, I will add a supplemental analysis on this topic as largely guided by my literature review.

classification of compensation and reward attributes that will contextually improve the alignment between uniform (legacy) incentive structures and the group performance drivers.

The mergers and acquisitions (M&A) literature sheds some light on this issue. In order to successfully improve performances, acquirers typically offer substantial financial incentives to induce the managers to go above and beyond their normal job duties to champion aggressive organizational change and growth (Hitt, Harrison, & Ireland, 2001). While the unit of analysis is not a perfect match to that of this study's of radical groups' initiatives, the activity in these teams occurs within a separate 'unit,' or subset of the established organization that is new and different from the core organization. The effects of this practice are supported by empirical studies on compensation, which, in general, report a positive influence of monetary incentives on employee and firm performance (Booth & Frank, 1999; Gagne & Forest, 2008; Gerhart & Rynes, 2003; Lazear, 2000). Unfortunately, the exceptionally high rates of M&A failures indicate that increasing performance incentives do not consistently increase the performance of managers operating in such settings.

Support for a competing argument comes from social psychology theories such as Self-Determination Theory (Gagne & Deci, 2005) and economic theories such as Motivation Crowding Theory (Frey & Jegen, 2001; James, 2005), which assert that financial incentives can also *reduce* motivation and performance. Clayton's (2014) groundbreaking study on the topic of inducing championing behaviors looked at (1) financial incentives, (2) organizational support, and (3) shared vision. With reference to financial incentives, the study points out that the most common and longstanding practice

is to increase financial performance incentives via some combination of stock options, profit sharing, gain sharing, or individual bonuses. With items reflecting both individual and organizational incentives as a percentage of compensation, Clayton's (2014) study revealed organizational incentives did not significantly affect championing behaviors directly or indirectly, and surprisingly found that individual performance incentives directly *reduced* championing behaviors, supporting O'Connor and McDermott (2004) finding that individual performance incentives create downside risk since failure rates are high. Clayton's work is also supported by Osland, Kolb, Rubin, and Turner's (2006) work in which 'excessive focus on extrinsic motivation, is identified as a 'creativity killer.'

While studies about the impacts of financial incentives have been abundant, these studies signal significant gaps in how such incentive models apply for radical innovation and corporate entrepreneurship performance. Given inconclusive findings and that these studies are solely focused on financial incentives, they fall short of identifying combined effects financial and non-financial features that are missing from studies of radical innovation and performance of entrepreneurial groups. In short, the aforementioned research adds further question to the debate as to whether incentives/rewards matter and, if so, which ones matter.

The qualitative work by O'Connor, reviewed in groups A of the literature on radical innovation, asserts strongly on the side that the right (non-standard) incentives matter. Sadly, little is offered to practitioners and scholars of CE and innovation as to which combinations of non-standard incentive and compensation mechanisms foster performance and radical innovation. One place to find creative, non-standard incentives

would be to look at what employees are customizing for their unique situations. Specifically, analysis of idiosyncratic deals (or i-deals) which are mutually beneficial, reviews of personalized agreements of a nonstandard nature that are negotiated between individual employees and their employers would be a good start. This trend addresses realities of fluid organizational boundaries and heterogeneous outcomes sought by firms, which seems to add a mounting pressure on corporations to match compensation and incentive systems that differ from their legacy solutions. Where it is not feasible for corporate entrepreneurs to be appropriately compensated for their skills and tacit knowledge, alternative governance arrangements need to be explored (Phan et al., 2009).

An example of a customizable alternative incentive programs is evidenced by firms' openness to negotiating employment conditions to suit the needs and preferences of highly valued employees (Rousseau, 2001; Shore et al., 2004). I-deals are "voluntary, personalized agreements of a nonstandard nature negotiated between individual employers and their employees regarding terms that benefit each party" (Rosen et al., 2013: 710). Rousseau and colleagues (Rousseau, 2001; Rousseau & Kim, 2006) proposed that i-deals are negotiated across many diverse resources that workers value and that i-deals can be used to attract, retain, and motivate employees. Some empirical research supports this claim (see Hornung, Rousseau, & Glaser, 2009; Hornung, Rousseau, Glaser, Angerer, & Weigel, 2010; Rousseau, Hornung, & Kim, 2009) and has shown that (a) employees negotiate i-deals that are both extrinsic (e.g., compensation) and intrinsic (e.g., flexibility of work hours) to their work, and (b) successful negotiation of i-deals relates to outcomes such as heightened organizational commitment, initiative, and work engagement.

As a result, I will focus on specific compensation content explored in these studies, as well as those reported by O'Connor and McDermott (2004). These attributes such as access to talented people inside the company, company reputation, and prestige, which opens doors in new markets, or access to financial resources. Other features can be added (e.g., protection from downside risk, career progression support) based on my experience. These examples of incentives can be added to other standard features, all in support of expanding our understanding of the role of specific non-standard compensation/incentives in shaping the performance of innovative intrapreneurial teams. Additionally, I also found it important to inquire whether the compensation/reward model used for team members was the same model used for employees in the 'core' business.

Putting It All Together: Multi-Level Model of Antecedents of Intrapreneurial Team Performance

The small group (team) literature defines external activities as task-related activities that are directed toward the team's environment to manage its relationships with external actors, including other units within the same organization, such as with senior managers. This research defines internal activities as those intragroup processes occurring within the group boundary, including the forming and enforcing group norms, communication among members..." (Choi, 2002: 184–185). I will use Choi's distinction to assess and organize my antecedents described in multi-layered fashion from within, at the boundary, and then outside the team as to affect the team performance. Later, I will use these principles to discuss how to manipulate these drivers in practice in a proper, balanced way.

I adopted specific antecedents because of their fitness to account for the identified problem of practice. Their effects are also in need testing, per the literature review, for

various reasons. Specifically, as discussed above, they have been validated in the past with mixed findings (rewards/incentives), with unclear units of analyses (boundary spanning), remain untested in the specific study context (senior management relationship/communication); or have been left out entirely in the past studies in a similar context (boundary buffering, and IT capabilities). I layer these constructs by following an ‘inside-out logic’ to account for the position of each introduced construct in the nomological network, as depicted in Figure 2. The layers are as follows: (1) Intragroup (i.e., within group—team, unit) mechanisms including relational climate; (2) boundary mechanisms (boundary buffering and spanning); (3) environmental mechanisms including senior management relationships/communication; incentives/rewards, and IT capability); and (4) contextual influences which comprise of typical controls such as group age, autonomy, and company size. I will use this logic and lens to discuss these drivers of RI and the performance of entrepreneurial groups.

Putting It All Together: Outcome Variables

Our literature review of both ambidexterity and CE helps position the dependent variables of this thesis. The dependent variable, as mentioned in our CE literature review, commonly uses “innovation,” and perpetuates a product focus, not unlike the innovation literature, which is ill-suited for our research. This measure consists of items such as 1) number of new products or services developed, (2) number of new products or services brought to market, (3) speed with which new products or services are developed, (4) speed with which new products or services are brought to market...” (Goodale et al., 2011: 123). This product-centered measure, which lacks the more general scope of

“entrepreneurship,” clearly does not apply to the kind of RI of my dissertation, as outlined in Table C, which by design extends beyond product innovation.

Since the majority of studies in the ambidexterity stream are viewed from the perspective of the overall organization, studies have treated ambidexterity as a mix of incremental and non-incremental innovation, used as a dependent variable (i.e., Hill & Birkinshaw, 2014; O'Reilly & Tushman, 2008) and as endogenous variables (Gibson & Birkinshaw, 2004). Literature has also evolved into incorporating specific varieties of ambidexterity—contextual and structural (i.e., Zaidi et al., 2015), separately but as drivers of new product development.

One study (Tushman et al., 2010) examines business units; however, the study shares the same outcome measure of other studies that focusses on a mixed dependent variable, therefore is not suitable for our setting since striving to be ambidextrous, adept at the simultaneous exploitation and exploration, is different from our (groups') desired outcomes. Another exception, which is useful for our purposes, comes from the research by Gibson and Birkinshaw (2004), *The Antecedents, Consequences, and Mediating Role of Organizational Ambidexterity*, which examined ambidexterity as mediators on *business unit performance*. Borrowing from the team performance literature, we tap Hackman and Edmondson's construct for the use of their team performance scale, as well as two items from the performance as used by Gibson and Birkinshaw (2004). Perceived team performance is necessary since traditional measures (sales, profits) may require time to come to fruition for groups not yet matured. Ironically, even new venture team Klotz, Hmieleski, Bradley, and Busenitz (2014) reveal outcome measures relating to venture sales, number of employees, and firm profitability are about as common in NVT research

as in the general entrepreneurship literature. Indeed, the three most commonly used outcome variables used in the studies reviewed are growth in sales (34%), profitability (32%), and number of employees (15%). Many studies in the NVT literature have used growth (primarily in terms of sales and employment) as a proxy for firm performance, and few studies have considered the goals which may be different from the more quantifiable measures discussed. Secondly, in contrast to the pursuit of ambidexterity, we focus on exploratory innovation as the outcome. So in addition to the dependent variables of group performance, we adopt group radical innovation. Radical innovation captures the heterogeneity of the innovation envisioned in this thesis. The selected constructs for our model and analyses will be further defined and justified in my measures section.

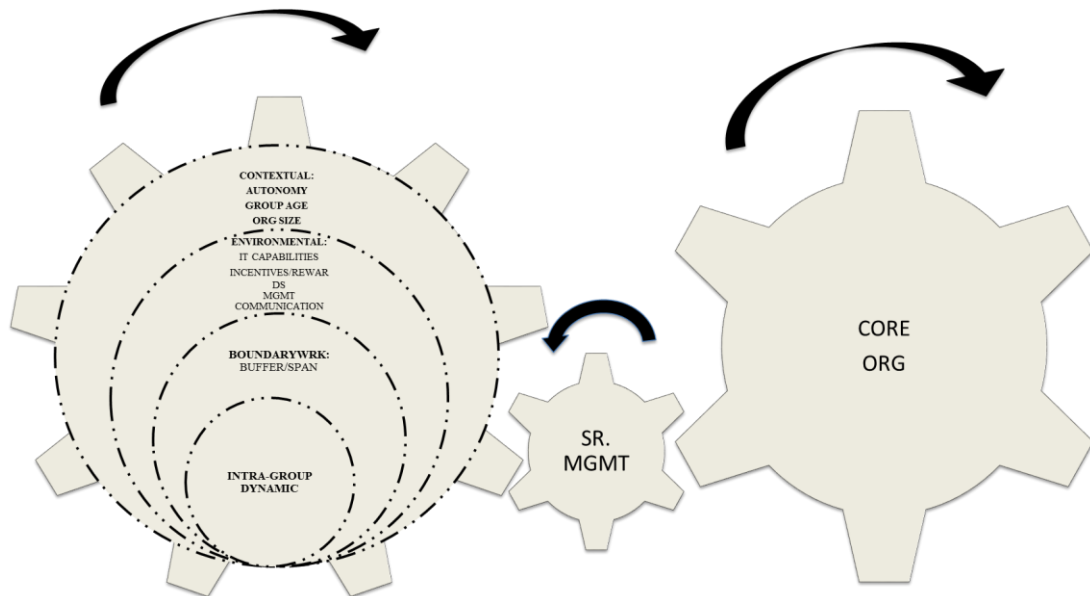
RESEARCH QUESTION AND RESEARCH GOALS

As discussed above, this study aims to build on and fill gaps in the existing literature on corporate entrepreneurship by introducing new ways to consider intra-organizational mechanisms that influence innovation and performance of entrepreneurial groups in established organizations. Above, we discussed the inside-out logic of our analysis in the illustration below in Figure 3 and followed by an addition to the wheels in orbit metaphor, which illustrates the wheel in the larger ‘system,’ Figure 4.

Figure 3: Inside-Out Logic of Variables



Figure 4: 'Wheels in Orbit' Part II: Radical Innovation Group Within Larger System



We chose the ‘group’ (or team) as the unit of analysis for practical reasons that ideate, plan, and execute the change. To this end, the study assesses first the impact of intra-group traits, especially the relational climate within the venturing group. We and examine the group’s empathy, energy, and resonance as indicators of intra-group change mechanisms. Next, we examine how the groups managed their boundaries using a nuanced analysis of boundary spanning and buffering. Other specific intra-organizational issues shaping the group outcomes include upper management communication/relationship, incentives, IT capabilities, and organizational drivers. Finally, the prevalence of legacy rewards and incentives will be determined and complemented by data gathered about types of rewards/incentives in place and preferred by entrepreneurial group members, through a supplemental analysis. Furthermore, a model and recommendations for the practical implementation of group initiatives are developed.

Figure 5: Summary of Research Problem, Question, & Goals

Problem of Practice	Unfulfilled potential of entrepreneurial initiatives inside established organizations.
Research Question	How to effectively manage entrepreneurial groups in established organizations tasked with pursuing radical innovation initiatives?
Research Goals	Goal #1: Identify and test the influence of key intra and extra group drivers (on radical innovation initiatives. (e.g., relational climate, boundary spanning and buffering, IT capabilities, and reward mechanisms) Goal #2: Develop and discuss of a model and recommendations for sr. managers and leaders of entrepreneurial groups within established organizations. Goal #3: Assess the prevalence and importance of specific non-standard incentives to group members, in order to fuel further study.

Research Goals

Figure 5 outlines the research goals in the context of the problem of practice and the larger research question.

This dissertation's primary goal is to identify and assess the influence of a select set of drivers of entrepreneurial initiatives on performance and radical innovation, while secondarily contributing to a model of sustainable corporate entrepreneurship. In fulfilling the first goal, a number of select drivers will be examined to determine their influence on outcome variables (performance, radical innovation). Specific questions that will be addressed include: Does within-group relational influence positively influence radical innovation and performance? Does boundary buffering influence radical innovation and performance? Does boundary work link with relational climate to mediate the influence on outcomes? Once the inner workings of the team are established, and the influences of managing boundary mechanisms around the teams are assessed, we proceed to explore the influence of environmental drivers within the organization on the team's radical innovation performance, including senior management, incentives and rewards, and IT capability. To do so, we address to what extent does (1) upper management's communication/relationship quality influence radical innovation and performance? Does a proactive IT stance have a positive association with radical innovation and performance? And, do performance-based rewards influence radical innovation?

The second research goal seeks to develop and discuss (for group managers and other senior managers) a model for senior managers regarding the implementation of entrepreneurial groups within the organization.

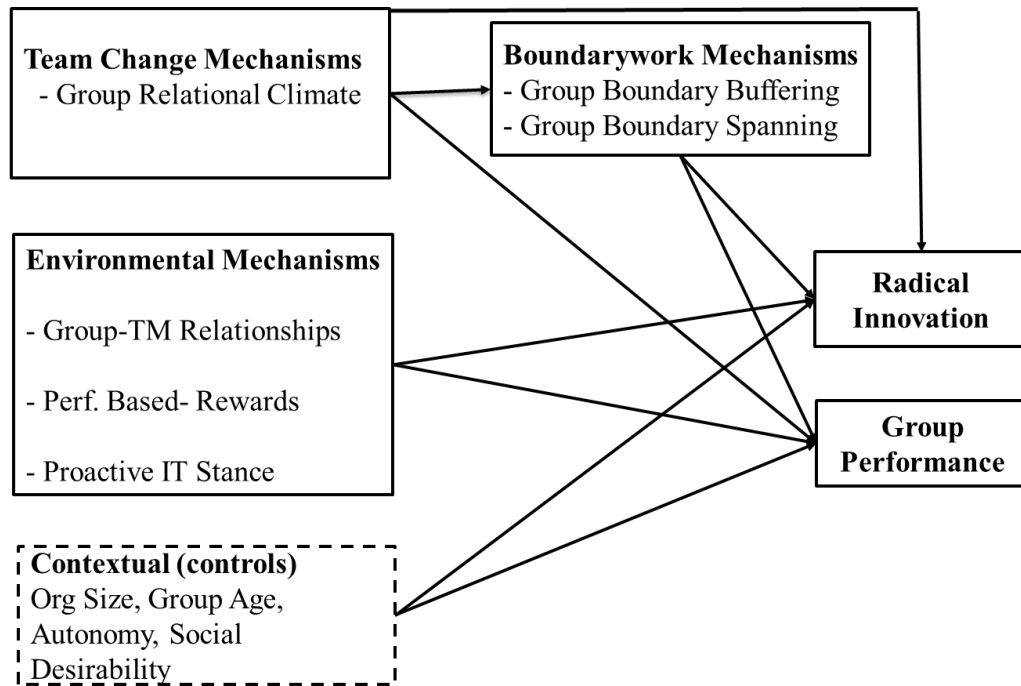
Third, we strive to delve deeper into rewards and incentives by assessing the prevalence of unique incentives vis-à-vis standard incentives of employees of the established organization and to identify incentives (that could fuel future studies) that align desired outcomes.

CONCEPTUAL MODEL & HYPOTHESES

Hypothesized Model

Based on the above literature review and research goals, our research model with hypothesized relationships is included below in Figure 6, and a summary of hypotheses can be found in Appendix A.

Figure 6: Hypothesized Model



Hypotheses of Direct Effects

Relational climate – intra-group mechanisms of change. This is not the first study to incorporate relational measures in studies of corporate entrepreneurship and

innovation; however, I opt for a construct not previously used in research of this kind, called *relational climate*. Dimensions of team climates relating to constructs of relational quality, shared vision, empathy, and ‘psychological safety’ as they relate to product innovation and team performance have emerged as variables used separately in the last couple of decades as antecedents or moderators, with mixed results. I find this exemplified in studies where employees are willing to take risks, which requires a sense of psychological safety that grows as relationships develop (Baer & Frese, 2003; Edmondson, 1999). Trust supports the efficacy of knowledge transfer (Levin & Cross, 2004), and trust develops and deepens as interpersonal relationships develop within the organization (Lewicki & Bunker, 1996). Through this study, I advance relational climate, a dimension of organizational climate conceptualized by Rochford and Boyatzis (2016). Arguing that high-quality relationships positively impact product innovation (Thompson & Heron, 2006) and that high levels of shared vision and compassion result in more innovative performance (Anderson & West, 1998; Bain, Mann, & Pirola-Merlo, 2001; West, 1990), Kendall found support demonstrating that the organization's relational climate, as perceived by individuals has a positive direct effect on product innovation success (Kendall, 2016).

Within a group, an individual’s perception of relational quality includes shared understanding, an atmosphere that is vibrant and energetic, and a feeling of mutual commitment to one another (Cohen & Prusak, 2001; Cohen & Bailey, 1997). Relational climate arguably goes further in explaining innovation outcomes. High-quality relationships are shared goals and knowledge, where a sense of deep mutual respect creates interconnections between people with affirming social interactions and a dynamic

social context such that people feel valued and appreciated, and can participate actively in the organization while being more fully themselves, and saying what they think and feel (Carmeli, Brueller, & Dutton, 2009; Dutton, 2007; Dutton & Heaphy, 2003; Dutton, Worline, Frost, & Lilius, 2006; Kahn, 1998, 2007). In short, tolerance for risk-taking, which contributes to the overall explained variance of intrapreneurship (Yariv & Galit, 2017), the overall need to create change outside the framework of the existing business (Sandberg, 1992), and innovation requires what is argued here to be represented by relational climate. I hypothesize, a direct path and from the more general relational climate construct, as follows:

Hypothesis 1. Group relational climate has a direct positive influence on 1a) radical innovation and 1b) performance.

Boundary spanning. In view of blurring formal structural control mechanisms, and to capture informal networks versus formal structures, two dimensions of boundary work—group boundary spanning and boundary buffering—are the focus. Boundary spanning refers to actions a focal team undertakes to reach out into its environment to obtain important resources and support (Faraj & Yan, 2009). The importance of boundary spanning comes from a prevalent theme in the strategic management and innovation literature supporting the assertion that firms increasingly need to rely on external knowledge sources to gain and sustain competitive advantage (e.g., Chesbrough, 2003; Dyer & Singh, 1998; Teece, 2000). Relevant research also refers to organizational practices that leverage and absorb customer knowledge that improve innovation performance (Foss, Laursen, & Pedersen, 2011). In the corporate entrepreneurship domain, Glaser et al. (2015) found a positive direct effect of top management boundary-spanning on exploratory innovation, As described in Yukl's research in the late 1970s

and early 1980s, descriptive research on managers found that it is important to influence bosses, peers, and outsiders as well as subordinates (Kaplan, 1984; Kotter, 1982; Mintzberg, 1973), and later research on teams found that boundary-spanning behavior is important for effective team performance (e.g., Ancona & Caldwell, 1992; Joshi, Pandey, & Han, 2009). As Faraj and Yan (2009) show in their research on boundary work in knowledge teams, boundary spanning remains an important variable with the emergence of post-bureaucratic organization, in which teams are a primary organizing structure. Reviewing the existing stream of team boundary spanning literature across multiple levels of analysis, Marrone (2010) establishes a relationship between team boundary spanning and team outcomes such as innovation and performance. For these reasons, I hypothesize as follows:

Hypothesis 2. Group boundary spanning has a direct positive influence on 2a) radical innovation and 2b) performance.

Boundary buffering. Formal structures and boundary spanning as they relate to radical innovation had been focused in the extant literature (i.e., Glaser et al., 2015; Raisch et al., 2017; Rossi, Festa, Papa, & Scorrano, 2019) but there is a paucity of literature espousing the importance of buffering in the context of breakthrough innovation in established organizations.

Boundary buffering is defined as an action that protects core processes, may be undertaken either in response to or in anticipation of disruptive forces within the environment. As teams increasingly become central work structures, it is becoming necessary to devise ways to ensure that pivotal work functions are accomplished, and team members are protected from external demands. Therefore, another aspect of boundary work, as raised by Faraj and Yan (2009), is adopted by our study and relates to

boundary buffering. Both boundary spanning and boundary buffering variables are important to examine in the proposed model.

The management of boundaries within and outside the group seems paramount and inevitable, given the importance that is placed on entrepreneurial groups for intrapreneurship to flourish amid the external trends driving internal organizational environments, as discussed.

The chief purpose of boundary buffering (Faraj & Yan, 2009), is noise reduction. Without buffering from internal distractions, it becomes easy to imagine that without insulation from the internal, inward-looking focus on internal headwinds, that externally focused perspective becomes undermined. Given the importance of insulating the unit's activity from intra-organizational interference, it seems plausible that boundary buffering, conceptualized as one of the types boundary work, is particularly telling for this research acknowledging the countervailing forces at work in large organizations that work counter to the performance and intrapreneurial behavior required of the non-incremental innovation-seeking unit within the organization.

We believe that buffering is an important positive influence of radical innovation team outcomes, therefore positing a paradoxical claim since we also hypothesize the positive influence of boundary spanning. Raisch et al.'s (2017) research on organizational boundaries found that activities focused on the creation and reinforcement of boundaries need to be combined with boundary-spanning activities (Ancona & Caldwell, 1992; Miller, Fern, & Cardinal, 2007). Likewise, we resolve this paradox in arguing that the 'opening' and 'closing' of the team to outside influences are both important, and we argue that managing the 'doors' call for the careful orchestration and balancing of

opening the group to outside resources and knowledge (spanning), and closing, or blocking the group from distraction (buffering) and contrary exploitative activities of the ‘old’ business.

Hypothesis 3. Group boundary buffering has a direct positive influence on 3a) radical innovation and 3b) performance.

Group-senior management relationships. Our model accounts not only for relational climate of the entrepreneurial team, as discussed, but also the relationship above it—the upper-level influence that connects the team to the existing business leadership and environment. The relationship I find most relevant to decisions affecting the eventual form of the venture unit is the one focused on the relationship between senior managers and the entrepreneurial group. This relationship is critical since innovation strategies are shaped at the top management level. This constitutes a key aspect of the problem of practice relating to decisions affecting the organizational form, and the very fate, of the entrepreneurial group. This can lead to dissolution or integration of venture, not to mention spin-outs or spin-ins. A top-level support and productive relationship within the venture can be for naught, if not devastating, should a disconnected senior leader make uninformed strategic decisions on behalf of the venturing team. On the other hand, a senior leader who has a good rapport, including frequent communication, could be there to understand and support the venturing team when needed. In this scenario, the leader can appreciate the uniqueness, strengths, and weaknesses of the unit as strategic decisions with regard to the group’s eventual fate are made.

To follow a sports metaphor, Boyatzis points out that the team's coaches, owners, and general managers play all critical roles in linking the team. They link the team to upper echelons of the organization and intermediate the team's effort between the organization and the community, managing public relations, fans, and the political community (Boyatzis, 2010). In their study of business units in global companies during exploratory innovation, Glaser et al.'s (2015) findings point to strategic renewal being a top-down phenomenon, as TMT (Top Management Team) boundary-spanning has a positive effect on unit's exploration. Upper management support for idea generation has been used in studies of entrepreneurship (Damanpour, 1991; Hornsby, Naffziger, Kuratko, & Montagno, 1993; Kanter, 1996; Pinchot, 1985; Stevenson & Jarillo, 1990; Sundbo, 1999), which emphasize the role played by senior managers in corporate entrepreneurship and innovation by encouraging innovators to 'bend rules and rigid procedures in order to keep promising ideas on track.' While the DV was ambidexterity by Hill and Birkinshaw (2014), the setting examined venturing units relationships with upper management, and found that the direct effects between a positive relationship and ambidexterity were positive and significant. As with other researchers who argue that the good relationships are essential (van den Bosch & Duysters, 2014), the following hypotheses posit a positive influence of relationship with senior managers (to whom the initiative reports directly to and to those outside of the direct reporting relationship) as having positive effects on the outcomes. Based on the foregoing, I posit:

Hypothesis 4. Positive Top Management relationships (between the initiatives' leader(s) and parent leaders) positively influence 4a) radical innovation and 4b) performance.

A competing view of the aforementioned hypotheses would argue that senior management intervention does not result in a positive influence on radical innovation and group performance. Senior leaders understandably are more used to managing exploitation, and there the venturing team should be physically, mentally, and organizationally separated from the main organization (e.g., van den Bosch & Duysters, 2014). As harsh as it may sound, frequent communication with exploratory groups may *not* be helpful, even if the intention is to be helpful. It may increase control of the group's pursuits, thereby reducing exploratory variance, when instead, the group seeks variance and needs slack in its pursuits to run 'contrary' in its ways which, by definition, are viewed as disruptions in that they depart from the customary ways of doing business. This view seems consonant with the research about knowledge flows influences on exploitation versus exploration activities, as discussed in our literature review. Specifically, there is a positive relationship between top-down knowledge inflows and *exploitation*. Mom et al. (2007) found empirical evidence that bottom-up and lateral knowledge inflows are positively related to a manager's *exploration* activities. The author further found that when the top-down knowledge inflow increases within the organization without enabling bottom-up or horizontal inflows, it is likely that managers increase their exploitation activities without increasing their exploration activities. This would lead to exploitation driving out exploration (Mom et al., 2007), leading to a negative influence of communication with senior managers on radical innovation and performance.

This 'stretch' hypothesis would seem to run counter to the findings of Hill & Birkinshaw as described, however, may speak to the difference between the aggregate DV of ambidexterity versus a singular focus on exploration. Finally, a negative direct

effect may become more imaginable when also considering that in smaller organizations when compared to the core, the entrepreneurial group is easily overlooked and will be less powerful, require disproportionate resources, and have long term pay-offs. For these reasons, we hypothesize that the influence of the relationship with senior management will be negative on the entrepreneurial group's outcomes.

Hypothesis 5. Positive Top Management relationships (between the initiatives' leader(s) and parent leaders) negatively influence 5a) radical innovation and 5b) performance.

Reward (incentive) mechanisms. One critical area to assess is the role and impact of organizational level incentive systems that dictate how the team's members are evaluated in the context of the whole corporation. In their work on *Corporate entrepreneurship: Current research and future directions*, Phan et al. (2009) underscore unique challenges that compensation places during entrepreneurial ventures. They argue that questions need to be asked from team members how they are and should be compensated and whether internal equity norms can or should be set aside to foster higher risk taking. The authors assert that such questions are also unique to corporate entrepreneurship (because most corporations have established organizational procedures for compensation) and not entrepreneurship in general. They further posit the need for asking fresh and distinct questions around this phenomenon, rather than attempting to apply accepted principles from the rest of the corporate context. The resistance to customize incentive systems may high and hard to overcome, a conflict conveyed already by General Electric's Jack Welch, who once said, "We can't have people in separate rowboats," in reference to a venture team's incentives. "We don't want anybody in our company going to a meeting with a different interest from everybody else" (van den

Bosch & Duysters, 2014). In short, corporate leaders are troubled by the need for the disparity between compensation and incentives among business units and radical business ventures.

Nevertheless, Hisrich and Kearney (2014) discuss in *Compensation and Incentives Factors for Corporate Entrepreneurship*, that there should be both financial and nonfinancial compensation and incentive plans for corporate ventures, including equity, bonuses, salary increases, career progression and advancement, and recognition and rewards. If the management tries to convince the employees to act like intrapreneurs, it must also be willing to pay them as entrepreneurs (Thornberry, 2003). If the employees have a high level of trust in the reward systems, they will be more willing to accept that organizational success will turn to be beneficial to all parties. Consequently, both their commitment to pursuing innovation (e.g., Bulut & Alpan, 2006; Chandler, Keller, & Lyon, 2000; Morrison & Robinson, 1997) and their willingness to assume the risks associated with it (e.g., Kuratko et al., 1990) will also be higher.

On the surface, it is obvious that a performance-based reward system should be part of creating a suitable internal environment for corporate venturing activities (Fry, 1987; Hornsby et al., 2002; Souder, 1981). Sadly, a 6-year longitudinal study of 12 radical innovation projects in 10 large established U.S.-based firms found that reward systems customized to encourage radical innovation in intrapreneurial units are rare, and “documented the strong mismatch that occurs between risks and rewards for radical innovation team members. In all but one firm, radical innovation teams were compensated no differently from anyone else in the firm” (O’Connor & McDermott, 2004: 25). The study further highlights that the downside risk for engaging in radical

innovation was high, in that only in one instance, there was an upside reward in terms of shared equity in the venture. In general, the study observed that there is a mismatch in risks required RI team members to take and the reward mechanisms put in place (O'Connor & McDermott, 2004).

My literature review also revealed a lack of pertinent research with regards to varying compensation/incentives and how they relate to innovation and performance outcomes. While the literature illustrates a wide variety of impactful entrepreneurial factors, there are few elements that remain consistent throughout the studies in the field. One appears to be an appropriate use of rewards (Block & Ornati, 1987; Fry, 1987; Kanter, 1985; Sathe, 1985; Scanlan, 1981; Souder, 1981). Researchers stress that in order to make a reward effective, it must encompass goals, feedback, emphasis on individual responsibility, and connect rewards with results. As for empirical work, there is limited and mixed evidence that specific reward systems have a strong effect on individuals' tendency to behave in an entrepreneurial manner (Block & Ornati, 1987; Sathe, 1989; Sykes, 1986). Kuratko et al. (1990) identified "reward and resource availability" as a principal determinant of entrepreneurial behavior by middle-and first-level managers. Similar results have been reported in subsequent studies (e.g., Hornsby et al., 1999, 2002; Morris & Jones, 1999). However, in testing for the presence of such effects in a large sample of senior executives (n=177), researchers (Goodale et al., 2011) found *no* support that rewards/reinforcements affect innovation performance.

Furthermore, in a study that considers impacts of organizational support dimensions on an individual's innovative performance, Alpkan et al. (2010) found partial, rather than full support of the relationship, specifically finding that performance-based

reward system significantly correlates with innovative performance. However, they were ineffective in regressing together the two significant drivers of innovativeness, namely support and tolerance. One reason for the mixed findings could be that researchers rarely differentiate between members who serve on entrepreneurial initiatives and members who are working on ‘core’ business pursuits, as discussed in the literature review.

Notwithstanding the mixed findings about the influence of performance-based rewards on ‘innovativeness,’ I will hypothesize that rewards mechanisms *do* make a difference in our sample on radical innovation. Specifically, defined as the ‘availability of a performance system based on reward system encouraging innovativeness,’ performance-based reward system will be tested in our context of entrepreneurial groups and radical innovation, as in the following hypothesis:

Hypothesis 6. Performance-Based Reward System (for innovativeness) positively influences radical innovation.

Proactive information technology (IT) stance. Ireland et al. (2009) claim that technological change creates a situation where competitive advantage is short-lived. Barring strong appropriability regimes that protect technological knowledge, firms with technology-based competitive advantages now must scramble to exploit the innovations before forces of fast technological diffusion and price competition erode them (Teece, 1986). Ireland, Covin, and Hornsby (2008) liken firms in such environments to being on an ‘innovation treadmill.’ Firms must continuously innovate to stay in the game, and they must innovate ahead of their competitors—in ways or in arenas that markets value—to achieve competitive superiority (Ireland et al., 2008). As observed by Lu and Ramamurthy (2018), the literature regarding IT capabilities specifically is divided when it comes to its influence on innovation outcome variables. On the one hand, IT capability

provides the ‘wiring’ for adaptive enterprise by building new digital options (Bharadwaj, El Sawy, Pavlou, & Venkatraman, 2013). On the other hand, IT can hinder organizational performance and innovation when firms are held back because of technological inertia created by large, complex, and inflexible technologies often expressed by the level of technical debt. In this study, we adopt a dimension of using IT capabilities for innovation called Proactive IT and advance the position that a proactive IT works in favor of radical innovation and performance. A proactive IT stance characterizes a firm that always searches for ways to explore or exploit its IT resources to create and capitalize on business opportunities (Lu & Ramamurthy, 2018: 937). A proactive IT stance has specifically been found to allow firms to spot and act fast on market opportunities (Zaheer & Zaheer, 1997). This enables learning and renewal, and by doing so, IT becomes a proactive partner in the innovation process and helps dynamic reconfiguring in changing environments. In short, the firm with a proactive IT stance can enable the rapid execution of innovative, radical actions. Drawing on this background, I hypothesize the following:

Hypothesis 7. A Proactive IT stance positively influences 7a) radical innovation and 7b) performance.

Hypothesis of Mediation: Could Boundary Work Mediate Relational Climate?

We start from the assumption that that relational climate within an established organization is only a part of the innovation and performance equation. That is, the team’s (intragroup) qualities can only go so far in achieving results without considering its interaction at the boundary within the larger organization. Although the relationship has not been explored previously, we believe that boundary work variables serve to transfer effects of relational climate on the outcome variables i.e., the boundary work

based on relational climate affects the environment in ways that is conducive to positive outcomes. This explains more comprehensively the chain of causal effect that the internal relational climate has on radical innovation and group performance in that many of these effects how well the team manages its boundaries. Organizational teams seldom work free from external observation and seldom have all of the resources they need within their boundaries (Faraj & Yan, 2009: 606). They must have the ability to interact based on their vision and commitment beyond the team's boundary. In the case of boundary spanning, the 'close-knit' team, in combination with its reaching out into its environment to obtain important resources and support, is a tandem needed for teams to realize its vision and reach the radical innovation outcomes. Likewise, the team with higher relational climate can buffer its perspectives and knowledge, that is, 'act to protect core processes either in response to or in anticipation of disruptive forces within the environment' (Faraj & Yan, 2009), then it is better able to achieve its radical outcomes. Conversely, when the team is not able to buffer from the noise of the external environment, then the strengths and potential of the team climate will be diluted and outcomes not fully realized.

In short, we posit that the actions of opening and closing of the groups 'gates' to outside resources (and interruption) are the mechanism through which relational climate also influences radical innovation and performance, as follows:

Hypothesis 8. Boundary buffering positively mediates the relationship between relational climate and 8a) radical innovation and 8b) performance; and,

Hypothesis 9. Boundary spanning positively mediates the relationship between relational climate and 9a) radical innovation and 9b) performance.

RESEARCH DESIGN AND METHODS

Study Goals

A research method chosen for a study should match the research question (RQ). The overall research question—how to effectively manage entrepreneurial groups in established organizations tasked with pursuing radical innovation initiatives—was broken down into three research questions and goals discussed above.

In review, the first goal aims to identify and test the influence of key intra- and extra-group drivers on radical team performance. The hypothesized model and rationale presented were explained.

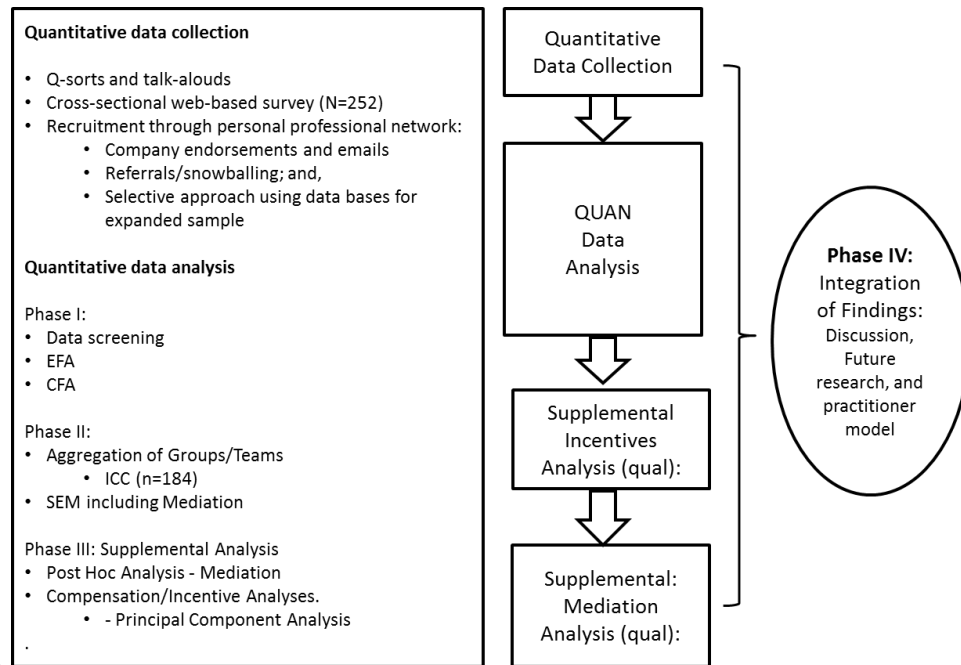
The second goal focuses on developing a model and recommendations useful to practitioners and will build on the ‘scaffolding’ that was started in our literature review. This will require integration of the results collected along with literature relating to multi-level analysis, as outlined in the theoretical framing of the literature that draws largely on dynamic capabilities and ambidexterity streams for a senior-level perspective of the group. This section can be considered an integration of literature, the quantitative portion of the study, along with non-quantitative information to gain an understanding of the data.

Thirdly, we delve further into ‘performance-based rewards’ and seek to indicate the prevalence of standard incentives, as well as to identify and note the importance of specific incentives. This part of the study explores the attributes of incentives/rewards that could be identified in order to narrow the mismatch between uniform (legacy) incentive structures and entrepreneurial group drivers. By definition, “qualitative research is about questioning the pre-given (taken for granted) variables, but it is also about

making new distinctions of any type of phenomenon” and is a way to get “closer to the phenomenon studied.” In this regard, this subsection of research incorporates a ‘qualitative dimension present in quantitative work’ (Aspers & Corte, 2019). Therefore, while the study overall is largely quantitative, a supplemental section of the survey asks questions about compensation and incentive plans. In this regard, this section is not driven by a goal to generalize, but rather to obtain insights into a phenomenon, individuals, or events for understanding the relating underlying phenomena Onwuegbuzie and Collins (2007).

In summary, priority is given to the quantitative methods in my study and follows a rationale for a “QUAN→qual = complement quantitative results” design. While not the priority strand, the qualitative supplemental strand will be guided by findings in the quantitative phase, anticipating complementarity attention to a variable (performance-based incentive systems) that has received mixed findings and is limited in specific types of incentives. The quantitative and non-quantitative information was collected from a single sample and, therefore, utilizes a mixed-methods concurrent sampling design (Onwuegbuzie & Collins, 2007: 293; Teddlie & Yu, 2007: 218). A flowchart with corresponding procedures of the entire research process is included in the side by side Figure 7.

Figure 7: Implementation Procedures and Flowchart of Research Design



Unit of Analysis

My unit of analysis is the entrepreneurial group pursuing radical innovation initiatives, as previously discussed. More specifically, it includes cross-functional groups participating in non-incremental initiatives within an established company or organization.

The sample's cross-functional groups operate "at the organizational boundary and stretches current organizational products and services, technologies, norms, orientations, structures, or operations into new directions" (Antoncic & Hisrich, 2003: 20), Zahra (2015) observed that there is an ongoing dispersion of corporate entrepreneurship activities, and with this in mind, we cast what we consider to be a wide net represented by cross-functional groups that capture the heterogeneity within organizations that promote radical innovation.

Measures

A summary of measures to be used for this study was provided in Appendix D and are discussed in this section. I expand in Figure 8, one that shows the specific constructs that correspond to the variables, including the level of analyses.

Figure 8: Inside-Out Logic of Variables, Levels of Analysis and Constructs

Inside 	Level	Mechanisms	Constructs	
			Exogenous Variables	Endogenous Variables
	GROUP	INTRA-GROUP	Relational Climate	-Exploratory Innovation -Performance
	GROUP	BOUNDARYWORK	Boundary Buffering Boundary Spanning	
	GROUP, ORG	ENVIRONMENTAL: - Mgmt/Sr. MGMT	Mgmt Relationships	
	INDIV, GROUP, ORG	- Incentives/rewards	Performance-Based Rewards	
	GROUP, ORG	- IT Capability	Proactive IT Stance	
Out	GROUP, ORG	CONTEXTUAL	Autonomy, org size, group age, social desirability	

A five-point Likert scale was used all latent variable constructs, which meant that two scales previously using seven-point scales (e.g., relational climate, radical innovation) were converted.

In addition to the prominent use of reflexive variables, the survey instrument contains a section of questions that are not designed to be reflective constructs for Structural Equation Modeling. Sample questions relating to incentives for the supplemental analysis, and are included in Appendix E.

Constructs that Reflect the Level of Analysis: As Hackman (2003) discusses, scholars generally conduct research at a single level of analysis (such as the individual, the group, or the organization level), although they often turn to the next-lower level for explanatory mechanisms. Constructs are designed as such, with survey design and instructions to differentiate individual, team, and organizational levels. The antecedents are at the group level, with openness to crossing levels in the case of IT (Proactive IT Stance) and more clearly crossing levels with regards to Relationships with senior management. In these two cases, the survey questions and constructs reflect the potential influence of the larger organization by asking respondents about the communication with senior managers that they report directly to as well as regarding communication with senior managers outside of the direct reporting line. Proactive IT Stance could span levels and is, therefore, more generically worded as to capture whatever level in the organization that drives IT Stance, whether it is the organization or group level.

Endogenous Variables

The group unit of analysis needs to be matched with the heterogeneity of outcomes that encompass group performance sought by groups pursuing non-incremental innovation at different ages of their development. The following two constructs for outcome variables were borrowed from a variety of literature streams.

Performance. Quantitative measures such as market share, revenue, and profits used in many studies are inadequate for capturing capture performance of the entrepreneurial group activities sought in this study, including those of newly formed groups or ‘ventures.’ Instead, we draw from an established Team Performance scale that functions for a variety of entrepreneurial activities and group life stages. The scale by

Hackman, as adopted by Edmondson (1999), includes: “this venture does superb work” 1; “this team meets or exceeds expectations”; and, “this team keeps getting better and better.” Additionally, two items from Gibson and Birkinshaw (2004) were adopted for their relevance and are expected load on the same factor in the Exploratory Factor Analysis (EFA). These items are “this business unit (group) is achieving its full potential” and “people at my level are satisfied with the level of this business unit’s (group’s) performance. The measure will be one of ‘perceived group performance.’

Exploratory (radical) innovation. Most traditional measures of innovation focus on product enhancements; however, this study’s focus is on a more varied non-incremental innovation that can be expected from autonomous and semi-autonomous venture entities. Exploratory innovation is defined as experimenting with new product–market combinations and requires building new skills and capabilities (Belderbos, Faems, Leten, & Looy, 2010; Benner & Tushman, 2003).

The exploratory innovation measure adapted by Jansen and Van Der Bosch (Glaser et al., 2015; Jansen, Van Den Bosch, & Volberda, 2006) matches the kind of strategic new business, market, product and service changes of interest, and will be used in this study. Examples of the six-item scale include “we experiment with new products and services in our local market” and “we commercialize products, and we frequently utilize new opportunities in new markets.”

Exogenous Variables

Intra-group mechanisms - Relational climate. The construct was developed by Rochford and Boyatzis (2016). Defined as a relatively stable collective structure that represents the socio-emotional atmosphere that is created and maintained as people

interact in a dyadic relationship, team, or organization, the construct represents the overall interpersonal atmosphere in an organization as a measurement of the quality of relationships. The 12 items, in three categories—compassion, relational energy, and shared vision—are concerned with how respondents view the climate in your group, and in particular, how people on the team interact with each other. Respondents were asked to indicate the response that best indicates the extent to which they agree or disagree with the statements listed.

Boundary work mechanisms (boundary spanning & boundary buffering). Faraj and Yan's (2009) work confirmed the importance of team-level boundary work for knowledge task teams and included a new and distinct form of boundary work called boundary buffering. The outwardly looking boundary spanning is defined as the "extent to which the team encourages its members to solicit critical information and resources, influence important external agents or actors, and build relational networks." Two of the four boundary spanning items are (1) To what extent does the group encourage its members to try to influence important actors elsewhere in and/or beyond the group on behalf of the group and its work? and (2) To what extent does the group value group members for making use of their relationships with others on behalf of the team?

Boundary buffering. Defined as the extent to which the team and its leadership deflect or absorb external interference and competing demands or requests and how well they prevent team members from overloading, boundary buffering construct consists of four items, and includes items such as "To what extent are outsiders prevented from "overloading" the group with either too much information or too many requests?; To what extent does the group or group leader help members manage demands placed on

them by other organizational units?; and To what extent do you feel that group members work in a well-buffered or protected environment.

Senior management relationship/communication. Borrowing from Hill and Birkinshaw's (2014) construct, respondents were asked to indicate the frequency with which you, or your group, communicates with the following individuals or other groups. In this case, the 5-point Likert scales consists of the following: 1 = never, 2 = rarely, 3 = monthly, 4 = weekly, 5 = daily. While it is an established tradition that holds that the frequency of interaction between ties is a proxy of the strength of their relationship, this research will remind the reader that the two terms are not identical, and substitutes "frequency of communication" as the name of the construct.

We selected the relationship/communication construct from Hill and Birkinshaw over CE's 'management support' construct (Alpkan et al., 2010; Damanpour, 1991; Hornsby et al., 1993; Kanter, 1996; Pinchot, 1985; Stevenson & Jarillo, 1990; Sundbo, 1999). Items included in this construct are: "Upper managers encourage my group's innovators to bend rules and rigid procedures in order to keep promising ideas on track," and "developing one's own ideas is encouraged by upper management for the improvement of the corporation." However, our talk-alouds confirmed the ambiguity over the unit of analysis and, more specifically, to whom 'upper management' referred. Instead, the 'Sr. Management Relationship' measure is clear about two specific sr. management positions (i.e., senior executives at corporate level your group directly reports to), allowing us to more precisely obtain insights about the hierarchical layer(s) above the group.

Proactive IT stance. A proactive IT stance characterizes a firm that always searches for ways to explore or exploit its IT resources to create and capitalize on business opportunities (Lu & Ramamurthy, 2018: 937). This measure of the same name Proactive IT Stance, comes from a dimension of an IT Capability construct, per Lu and Ramamurthy (2018), which contains four items and asks: “Relative to other firms in your industry, please evaluate your capability in acquiring, assimilating, transforming, and exploiting IT knowledge in the following areas...” Sample items include, ‘we constantly keep current with new information technology innovations,’ ‘we are capable of and continue to experiment with new IT as necessary’, and ‘we have a climate that is supportive of trying out new ways of using IT.’

Performance-based rewards. Defined as “availability of a performance-based reward system encouraging innovativeness” (Alpkan et al., 2010: 735), the construct known as *Performance Based Reward System for Innovativeness*, as used by Souder (1981), Fry (1987), Cissell (1987), Sykes and Block (1989), Kuratko, Ireland, Covin, and Hornsby (2005), includes the following items, among others: group members will be appreciated by their managers’ if they perform very well; group members from every level (of my unit) will be rewarded, if they innovate; group members with innovative and successful projects will be highly rewarded; and, group members with innovative and successful projects will be highly rewarded.

Contextual - Control Variables

We controlled for the following relationships that should be taken into account: organization size, autonomy, group age, and social desirability.

Team autonomy. Job autonomy is one of the core constructs in the assessment of climate for corporate entrepreneurship (Hornsby et al., 2002; Hornsby et al., 1993). Job autonomy has been demonstrated to correlate with the number of ideas implemented by higher-level managers (Hornsby, Kuratko, Shepherd, & Bott, 2009) and firm-level corporate entrepreneurship (Ling, Simsek, Lubatkin, & Veiga, 2008). Moreover, at the individual level, previous studies have connected job autonomy with various elements of intrapreneurial behavior. Defined as the ability to independently determine how to do a job or task, the construct has been empirically associated with innovative work behavior (Axtell et al., 2000; de Jong & den Hartog, 2005). The team autonomy scale was particularly attractive in that it was tested and used by Langfred (2005) with teams, and because of its high reliability. In this data set, it had a Cronbach's alpha of .91. For parsimony, six items were used in the survey, dropping three items with the lowest loadings or because of redundancy.

Company size (employees). Prior research has addressed firm size and is used as a control in this study based on the number of employees.

Group age. Previous theoretical and empirical literature relates the innovative behavior of firms according to venture age. Per Klotz, Hmieleski, Bradley, and Busenitz, (2014), the new venture can be broadly defined as a firm that is in its early stages of development and growth. The founding group of individuals is chiefly responsible for the strategic decision making and ongoing operations of a new venture. In general, such firms are in the process of bringing their initial products/services to market, forming a customer base, and putting into place organizational processes and procedures. Over

time, the corporate goals and venture goals and actors change, and new organization forms could take shape.

Social desirability. To obviate common method bias, a social desirability scale (Crowne & Marlowe, 1960) was incorporated into the survey. Six items from Crowne and Marlowe (1960) were used to measure and control for social desirability response specific bias.

Instrument development: Talk-alouds and Q-sorts. A pretest is a small, pilot study to determine how a questionnaire can be improved to minimize response errors (Converse & Presser, 1986), such as a respondent misinterpreting a question. Questionnaire pretests can be very important because response and other non-sampling errors are the major contributors to total survey error (Assael & Keon, 1982). More specifically, talk-alouds of the survey instrument were conducted with four expert corporate entrepreneurs. All ambiguous items were modified. Because of the within-group and extra-group relationships that are critical to examine in this study, particular attention was given to respondents' understanding of the distinction between the entrepreneurial group (within the established organization) and the rest of the company. The talk-alouds reinforced the need to regularly remind respondents of the distinction of the 'group' from the 'total' company, and definitions were developed. This important distinction was repeated with relevant question sets with instructions indicating whether the questions related to the group or total company, as follows: "When replying to the questions, please note that: 'group' refers to the entrepreneurial group that you're a part of (e.g., venture, cross-functional team, business unit, division), not the established total company. 'Total company' refers to the larger, established organization (e.g., corporate,

parent company, core business, its units/divisions/subsidiaries).” These adjustments allowed me to emphasize, identify, and change the vocabulary to minimize response errors and nonresponse errors (Bolton, 1993).

Q-sorts were also performed, which resulted in the removal or adapting of select items. These decisions are mentioned in the subsequent “measures” section. One item comprising the Relational Climate scale of 12 total items stating ‘our group’s daily work aligns with our vision’ was below my threshold in sorting results (of 53%) indicating lack of differentiation from other items, and calling attention to the ambiguity of whose vision—the group’s or the organization’s? Although the Q-sort raised these questions around the vision items of Relational Climate, I decided to include the remaining vision items in the EFA along with the other two dimensions—compassion and energy—staying aware of problems with the vision dimension of this construct. After changing the verbiage of ‘employees of my organization’ to ‘group members’ for consistency with the rest of the survey, the four items of Performance-Based Rewards were correctly matched at least 63% of the time and were, therefore, advanced to the survey instrument. One word was added to a Boundary Spanning item to clarify the question addresses *extra*, rather than *intra* team resources. The original item, “the team values team members for making use of their relationships with others on behalf of the team,” was changed to “the group values its members for making use of their *external* relationships with others on behalf of the team.”

Data Collection

The sample. The mid to large size company is chosen because the setting of complex organizations underscores the role of organizational climate in an era where the

changing competitive landscape is dictating that large companies become more entrepreneurial, innovative, and adaptive. Respondents served on autonomous and semi-autonomous, cross-functional teams pursuing change/innovative initiatives within an established (not new) organization that has more than 50+ employees (per Antoncic & Hisrich, 2001: 506). The age of the team/group could be new or established, and the age of the larger, overall organization (i.e., company, non-profit) should be 13+ years. With regard to group age, about half were less than five years old, and the remainder were five or more years old. Having recruited ‘cross-functional groups or teams, rather than business units working on breakthrough innovation initiatives, may explain what appears to be a youthful sample of groups, probably skewing the sample on the younger side vis-à-vis a sample of more mature ventures or business units.

With regards to managerial positions, I attempt to reach the top leadership of the team, as well as other upper-echelon leaders and managers in a variety of roles that include market-facing exposure, which is reflected in roles such as new sector development, business development, marketing, market development, strategic planning, business unit heads, and category managers/leaders. In short, since the goal is to examine a broad set of entrepreneurial and innovative initiatives, a deliberate attempt is made to avoid studying groups comprised solely of R&D roles or product (technology) innovations. Although these roles were not excluded, together, they comprised a small percentage of respondents at 10% of the sample.

Participants were recruited via two paths. The first was through my personal network of organization senior leaders. When contacts are provided by the established organization through cooperative arrangements, I requested company permission to

administer a survey to venture/business team leaders and to encourage participation.

Examples of organizations that submitted Letter of Collaboration at the present date (May 27, 2018) include a Fortune 50 company known for its entrepreneurship and innovation following the hire of its Chief Innovation Officer, my main contact, as well as a global labeling company, a credit union undergoing a transformation. The anonymity of individual responses was secured, and organization names, while requested voluntarily, are coded and not identified. To reduce the risk of common method bias, I attempted to get rankings of performance and innovation from organizational leaders who are familiar with these outcomes, although this proved difficult to ensure in most cases. Participants were also asked to identify other cross-functional members of their initiative within their organizations, as well as potential participants outside their organizations to receive the survey.

Respondents were further qualified by responding to whether they (1) currently serve on a cross-functional group within the organization pursuing innovative/entrepreneurial initiatives that seek to expand the business in new ways; (2) do not currently serve but have done so in the last three years; and (3) do not nor have not served on an initiative, in which case the survey is terminated. When I was not given access to contacts from organizations, I sought nested sampling design (Onwuegbuzie & Collins, 2007) by asking participants to recommend others who serve(d) on the initiative with them. In total, my personal network yielded 89 completed surveys of the 252 total. This data was collected between June 14 and January 15.

I also recruited participants by collaborating with Qualtrics to identify and recruit qualified participants. Because of the complexity associated with recruiting qualified

participants, several preparatory meetings were conducted to discuss appropriate screening and verification of participants to help assure the quality of the sample. I evaluated the responses in phases to ensure that the nature of the initiatives, as well as the diversity of roles and organizational settings were in line with the definition of corporate entrepreneurship as intended by this study's objectives. We targeted individuals who were involved in a team/group within their organization, seeking to expand the business in new ways (whether it is through new markets/segments, services, products, geographies, or business models). Positions were specific to mid-manager and higher levels in large organizations, in roles likely to be pursuing radical innovation, such as Market Development manager/director, Team Leaders; Innovation VPs/managers/directors/specialists; business development managers/directors; Strategic planning managers/directors; project managers/directors; Marketing managers/directors; Business Unit Vice Presidents/General Managers/Directors; New Sector managers/directors; International business GMs/directors; R&D managers/directors/VPs; engineers; and (new) Product Development/directors. Aside from those specific search criteria, secondary filters were imposed, and asked participants to declare (1) whether they worked for an organization established for at least 13 years, with over 50 employees; and (2) have participated on a team/group within the organization that pursues breakthrough growth/innovative initiative(s). Furthermore, participants were asked to describe the innovative initiative(s) that he/she served on, and that was being used as the reference for this survey. A total of 163 surveys of the 252 total number of surveys sample were qualified and completed along this recruitment path.

Additional Organization characteristics. Table 5 below includes a summary of respondent, group, and organization characteristics. For Company Activity, I grouped retail and wholesale distributors together under services, as per the service-providing industries ‘super-sector’ group of the United States Department of Labor. Educational institutions were also included under this ‘service-providing’ category. The largest subset of the service category comprised organizations in the healthcare sector, which, because of the size of this subgroup, is separated out in the sampling Table 5. Since there were seven remaining firms listed as diversified, I checked organizational web sites and other data collected (i.e., group activity) in order to slot these organizations in the proper group. Below are key characteristics of both the groups and overall organizations, assuming a sample size of 252 unless indicated. Concerning the age of the *organizations* in which the groups operated, 82% of the organization were at least 20 years old. Finally, it is worth pointing out that the vast majority of groups (94%) were part of organizations located and headquartered in the USA, serving the same (home) markets.

Table 5: Respondent, Group, and Organization Characteristics

Role on Group: Group Head/GM= 36% Business Dev/S&M = 15% Operations = 12% IT = 10% Prod Mgt/R&D = 10% Finance = 6% Other = 11%	Organization Size (Employees): <1,000 = 44% 1,001-10,000 = 27% 10,000+ = 28%	Organization Type (Sector): Manufacturing = 32% Services = 53% Healthcare = 12% Diversified = 3%
Organization Age: <19 Years = 18% 20-49 Years = 39% 50+ = 43%	Group Age: <5 Years = 49% 5+ Years = 51%	Group Compensation (n=236) Same: 77% Different: 23%
Group Head/GM (i.e., Group/Project Lead; Managing Director) Business Dev/S&M/Strategy (i.e., New Sector Director; Brand Dir; Key Acct Dir)		

Examples of Groups

The 252 respondents surveyed serve on 184 groups within 181 mid- to large-size organizations. I provide below descriptions of the cross-functional group initiatives as follows. The following vignettes provide a short summary of six of the groups' pursuits.

- A Fortune 500 Manufacturing Co., for a number of years, had grown through numerous acquisitions, but senior management recognized the need to develop strategies to leverage growth from within using existing resources. However, the company wanted to evolve to provide truly innovative solutions, not supply individual commodities. Instead of changing existing products, the company formed teams/units to help create a focus on innovation and entrepreneurship, providing entirely different products and systems that open and shape new markets, including the use of robotics for healthcare, an entirely new segment for the company
- A leader of training and therapeutic (healthcare) products saw its growth plateau in the early 2000s as competition intensified and the company rested on its laurels and became outdated. In an effort to modernize and regain its leadership reputation, and turn around the revenue decline, the company embarked on new initiatives that involved opening new markets segments, launching new services through new channels, and new business models involving digital channels.
- A leader in the increasingly automated packaging industry with clients in the production of food, pharma, beverage, personal care, and a variety of industrial goods. A strategic company leader recognized the need to break into new market segments. The new sector team pursues new, less saturated growth segments where the company holds little to no market share but meets a number of attractiveness criteria for mid and long-term growth. Although the growth initiative is not housed in a separate unit, the senior leader recognizes that cultivating clients in new segments is unique and cannot be handled by the established organizational sales and service organizations. Led by a VP of marketing, a semi-buffered (yet not dedicated) cross-functional team is assembled to provide the learn and serve new market segments until a stable growth trajectory is achieved.
- A large hospital system created a separate unit to launch into markets not currently competing in. These include entering new geographies and completely new services, not part of the hospital systems value proposition. The transformative and disruptive efforts include the launch of urgent care

facilities in new geographies that disrupt and fall outside conventional hospital settings.

- A for-profit institution of higher learning decides that it was missing out on an opportunity to serve cultural, global, and corporate markets. The Provost, a seasoned entrepreneur, set up a new 'divisions' of the company to create capabilities and serve these new underserved markets, which required new and unique services offerings, different from the conventional services provided by almost all functional areas of the company. While leadership changes led to questioning the need for unique service and separate division, the group stayed intact as the division prospered and in just three years, outperforming the established organization, of 100+ years, along with key financial and performance metrics.
- A financial services organization with \$2 billion in assets put together a team to launch the first all-digital comprehensive service organization in the United States. As it grows, it will be determined as to how/if the new and established businesses will be linked in the future; however, at least one organization leader predicts that the digital unit represents the future. In retrospect, the new business unit was launched, realizing its goal of offering digital-first seamless, personalized experiences has grown in the modern mobile world. Its strategic commitment paid off in remarkable ways by 'pivoting' and serving clients remotely during the pandemic. The new unit achieved 'aggressive growth, from \$0 to more than \$50 million in its first year.

Data Analysis

Pre-analysis data screening. The data were analyzed in two stages, as guided by the Flowchart of Research Design. Factor Analysis was conducted, first as an exploratory factor analysis (EFA), then confirmatory factor analysis (CFA), preceded by data screening.

As discussed in the sample discussion, respondents screened for organizations that are (a) both established and medium to large in size; and (b) growth initiatives within the organizations. Thirty-five were disqualified or incomplete, leaving a total of 253 records. Data collected from respondents who filled out the survey in the past tense were moved in SPSS to corresponding, present tense data fields. This included 12 records. Reverse coded item (SD6) was recoded.

After reviewing the data for unengaged responses, one case (R_vjhgXvwm6YEDjwd) was removed for responding *strongly agree* for every Likert scale. Ideally, the survey sought a sample based on various characteristics previously discussed. Among them, company size and age cutoffs were established. One case was removed for not hitting thresholds for being large or old enough. This left 252 cases to be included in the data set further analysis. Four records from two organizations had mis-recorded number of employees, which I corrected based on other respondents from the same organization and public information. Likewise, companies' ages were corrected based on known information in two cases. This left six respondents from two organizations, which fell below the ideal number of employees. These cases were retained based on the knowledge that the companies were in the IT or service sectors, and had recently embarked on aggressive outsourcing and/or relied on part-time employees, not reflected in the recorded data.

Missing data. A total of 14 values were missing, of which four were missing from reflective constructs and ten from categorical variables. In the former cases, I imputed with the median of nearby points for that construct. In the latter cases, the six missing number values of organization employees were lifted from other known information in the survey (colleagues who also filled out the survey).

For completeness, the variables were evaluated for skewness and kurtosis. There were no values that fell outside the acceptable skewness range of -2.0 and +2.0. Six structural model variables showed kurtosis issues for falling outside the same range; however, only two variables (RC1 of 3.29 and RC2 of 3.26) were outside the acceptable range of -3.0 and +3.0. I decided to keep them in the analysis at this stage. There seems to

be a lack of consensus regarding tests for normality and thresholds that determine departures from normality. Certain scholars consider numbers at 3.3 and above a red flag (Sposito, Hand, & Skarpness, 1983) while others more liberally consider threshold up to 10 (Kline, 2015). It is important to understand how the distribution departs from normality in terms of shape, and whether the values are large enough to warrant attention; however, given the small number and extent of the violating values and that Kurtosis affected Likert scale items, I decided to move forward with only a cautionary note concerning the two items mentioned; however, both items would be removed in subsequent analyses.

Normality. For completeness, the variables utilized in the structural model were evaluated for skewness and kurtosis. Left skewness was detected for seven items with values outside the acceptable range of -1.0 and +1.0. Kurtosis was detected for seven items (six of which also demonstrated left skewness) with values exceeding +1.

Homogeneity of variance. Because the data were collected in two phases, I conducted a test of homogeneity of variance comparing means of the sample groups for all the items in the factor analysis. The rule of thumb is that if at least one item per latent construct is invariant across the sample groups, then there is sufficient similarity. Indicates whether the participants responded to the questions in a similar way. Results indicate that all nine of the ten factors had at least one item with a mean that was invariant across groups, and of those, only one factor had less than one-half of the items with means that were invariant across groups. However, the means for Information Technology items had means that were significantly different across groups and, therefore, will be noted as a limitation of the study.

Exploratory factor analysis (EFA). An EFA was conducted, initially using Principal Axis Factoring as the method of extraction to determine unique variance among items and correlation to the anticipated factors. Promax rotation was utilized and selected due to the expectation of correlation among factors. Three of six Social Desirability items were also removed, as well as one of the Rewards items. The vision dimension of Relational Climate (four items) was removed for low loadings, confirming the lack of respondents' clarity of which vision (group or organization) signaled in Q-sorts and face validity.

The EFA was conducted to evaluate whether the observed variables loaded together as expected, demonstrated adequate correlation, and met criteria for validity and reliability. I then sought to achieve similar factor solutions across two extraction methods—Principal Axis Factoring and Maximum Likelihood—as an arbiter of top solutions. I first used eigenvalues to determine the number of factors with the expectation that there would be ten, which matched the result. For completeness, I followed the common practice of testing +1 and -1 fixed factors (testing nine and eleven factors). Neither test improved the pattern matrix. A different extraction was used (maximum likelihood) for comparison and reached a similar solution, with only minor differences concerning the loading of the two weakest Social Desirability (marker variable) items. By contrast, in the 'next best' model, to be used as an alternate in the CFA; however, items for one factor (Social Desirability) did not converge across two extractions, dropping the marker variable when the Maximum Likelihood rotation was used. Residuals were also higher in the alternate (5.0 versus 4.0).

Adequate explanation of 10 factors was obtained at 70.0% of variance explained, and 4% non-redundant residuals greater than 0.05. The lowest initial eigenvalue to be included in the model of 1.08 was clearly delineated from the next largest (eleventh factor) of .86. There were three of 45 items that had communalities <0.5 , but were retained for testing in a confirmatory factor analysis due to either approaching 0.5 or due to having only three items in a factor. KMO if .93 and Bartlett's test for sampling adequacy was also significant.

Convergent validity was met as evidenced by strong loadings, with the lowest of .48, well above the .35 threshold (Hair, Black, Babin, & Anderson, 2010), and discriminant validity was also achieved, as there were no significant cross-loadings (discriminant validity). No factor correlations exceeded 0.70. The pattern matrix is attached in Appendix A.

With one exception—the marker variable—all factors had good reliability measured with Cronbach's alpha; each was >0.70 , with the lowest of .78. All items are reflective of their constructs because they are generally interchangeable, internally consistent, and even small-item factors have high reliability as each item should adequately represent the latent factor (Jarvis, MacKenzie, & Podsakoff, 2003).

Confirmatory factor analysis. The final solution, consisting of a 45-item 10-factor model are as follows, which, as discussed, yielded similar factors and items (with only one minor exception involving a social desirability item) across both Maximum Likelihood and Principal Axis Factoring, was advanced for further analysis. Model fit was good and as follows: CMIN/DF = 1.59 with $p=0.0$; CFI = 0.924; SRMR = 0.049; RMSEA = 0.048; PCLOSE = 0.713).

Validity and reliability were checked, and per Table 6, showed convergent validity as evidenced by the AVE all above 0.50, reliability with CR all above 0.70. Discriminant validity is demonstrated by the square root of the AVE being greater than any inter-factor correlation. That is, comparing the maximum squared shared variance (MSV) of any inter-factor correlation to the average loading per factor, the latter average loadings are greater.

Table 6: Convergent and Discriminant Validity of the Constructs

	CR	AVE	MSV	RC	RAD	AUT	IT	PF	BB	BS	RW	SD	RL
RC	0.926	0.640	0.474	0.800									
RAD	0.888	0.533	0.463	0.636	0.730								
AUT	0.905	0.615	0.380	0.561	0.594	0.784							
IT	0.882	0.651	0.315	0.467	0.535	0.412	0.807						
PF	0.906	0.661	0.474	0.688	0.658	0.617	0.549	0.813					
BB	0.834	0.556	0.458	0.605	0.677	0.584	0.561	0.655	0.746				
BS	0.814	0.524	0.394	0.481	0.628	0.570	0.401	0.540	0.618	0.724			
RW	0.863	0.678	0.463	0.652	0.680	0.570	0.513	0.605	0.661	0.557	0.823		
SD*	0.661	0.401	0.266	0.516	0.452	0.422	0.316	0.489	0.352	0.419	0.436	0.633	
RL	0.784	0.646	0.134	0.147	0.242	0.135†	0.147	0.047	0.258	0.365	0.219	0.030	0.804

Common method bias. The issue of common method bias (CMB) was addressed. Because the data were collected largely using a single method, CMB, in this research, might come about from common rater effect (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). Nevertheless, it is important to point out that procedural steps were taken to minimize CMB as follows: extensive pre-testing of the survey instrument to refine survey instrument; order mixed criterion and predictor variables and blended most similar items, such as boundary spanning and buffering; using established scales; improving scale items; protecting anonymity; and, collecting third party data in some cases, if available.

In order to assess CMB, an unmeasured common latent factor was used first. Having established good model fit, I created and compared three nested models in which a CMV factor loadings were (1) freely estimated, in an ‘unconstrained’ model, (2) equal to zero, and (3) equal to each other. Comparing results of the nested models to assess the statistical significance of common method bias present in the data, the p-value was <0.05 for both equal weights and zero weights models, indicating that the models are rejected as meaningful models relative to the baseline model. Zero Constraints CMIN = 127.55; DF = 45; Equal Constraints CMIN = 44; DF = 44; with $p=0.0$. Assuming Equal constraints to be correct, the zero constraints model is: CMIN = 29.83; DF = 1 with $p= 0.0$. Therefore, common method bias is a concern. Secondly, a specific bias test was conducted using social desirability as the marker variable for which three nested models were compared. The results of this test were: Zero Constraints CMIN = 141.4; DF = 42; Equal Constraints CMIN = 66.8; DF = 41; with $p=0.007$. Assuming Equal constraints to be correct, the zero constraints model is: CMIN = 74.6; DF = 1 with $p= 0.009$. Figures of both tests in AMOS are included in Appendices B.

Although model fit remained very good in the bias-corrected models (CMIN/DF = 1.56 with $p=0.0$; CFI = 0.930; SRMR = 0.044; RMSEA = 0.047; PCLOSE = 0.832) for the Social Desirability marker; and (CMIN/DF = 1.522 with $p=0.0$; CFI = 0.935; SRMR = 0.040; RMSEA = 0.046; PCLOSE = 0.930 for the unmeasured latent marker variable), validities and estimates changed dramatically. Due to the disadvantages and the effect the common latent factor² and the effects of both this and the marker variable had on

² As cited by Grooms (2015), this may be consistent with disadvantages of using an unmeasured common latent factor (Podsakoff et al., 2003) such as: inability to identify the specific cause of method bias; inability to identify if the common method variance is “inflating, deflating, or having no effect on observed

measurement model validity and reliability, we chose to move forward with a Structural Equation Model (SEM) that does not include CMB-adjusted values in the SEM. Instead, Social Desirability was retained as a control variable.

Multicollinearity. I tested for multicollinearity by analyzing Variance Inflation Factor (VIF) in SPSS. VIF is a widely used measure of the degree of multi-collinearity of the independent variable with the other independent variables in a regression model. The Variable Inflation Factor (VIF) was tested for all independent (exogenous) variables resulting in an indication that these variables have little to no correlation with each other with all VIFs resulting <3.0 . By running regressions of all predictors on each of the two endogenous variables, radical innovation, and performance, and by observing VIFs for all predictors, I found none higher than 2.71, which is considerably lower than common thresholds, as indicated in Table 7. Variance inflation factors (VIFs) less than 10 are traditionally viewed as justification for a model's lack of multicollinearity, with 5.0 being ideal for reflective constructs (Cenfetelli & Bassellier, 2009). According to Hair et al. (2010), VIF of <5.0 is acceptable, and $VIF < 3.0$ is a good indicator that issues of multicollinearity are not present. Therefore, all VIF results in this model are <3.0 , and correlations are not high enough to cause concern for multi-collinearity.

relationships between predictors and criterion variables" (Podsakoff et al., 2003: 893). The factor may reflect not only different types of common method variance but also variance due to relationships between the constructs other than the one hypothesized" (Podsakoff et al., 2003: 894). Use of a common latent factor assumes the method factor does not interact with trait factors.

Table 7: VIF Multicollinearity Test (with R-Squares): Predictors on Each of Two Exogenous Variables

	Radical Innovation (R ² =63.1%)	Performance (R ² =62.7%)
Relational Climate	2.23	2.36
Information Technology	1.64	1.71
Boundary Buffering	2.33	2.44
Boundary Spanning	1.85	1.99
Relationships	1.19	1.20
Rewards	2.07	2.12
Social Desirability	1.45	1.45
Radical Innovation	N/A	2.71

Multivariate influentials. As a final step before beginning SEM, I examined Cook's distance multivariate analysis and ran linear regressions of predictor variables on each endogenous variable to identify any cases that might be distorting the regressions. As illustrated in Appendix C), I started with radical innovation and proceeded with the performance variable. Based on scatterplots graphs and Cook's distances, the most influential case(s) for each group were identified and noted on each of the graphs. Because the highest values were 0.13 and 0.21 on radical innovation and performance respectively, both well below a 1.0 threshold, I concluded that further examination of the data was unwarranted and that there were not any overly influential cases.

Together, the analyses of influential and multicollinearity provide confidence that cases are not over-influential in the data set, and that predictor variables contain acceptable levels of overlap in regard to the portion of variance they explain on the relationship to endogenous variables.

Table 8 contains the descriptive statistics and correlations among the variables. As with covariance structure models, it is essential to examine overall fit of the model, which

proved acceptable overall, with acceptable to excellent fit according to all indices, except RMSEA, as follows: CMIN/DF = 2.90 with $p=0.0$; CFI = 0.97; SRMR = 0.10; RMSEA = 0.087; PCLOSE = 0.012.

Table 8: Means, Standard Deviations, and Correlations

	<i>Variable</i>	<i>M</i>	<i>SD</i>	<i>RC</i>	<i>RAD</i>	<i>ATY</i>	<i>IT</i>	<i>PF</i>	<i>BB</i>	<i>BS</i>	<i>REW</i>	<i>REL</i>	<i>SD</i>	<i>Age</i>	<i>Size</i>
1.	RelClimate	4.11	.93	----											
2.	Rad Innov	3.83	1.11	.65**	----										
3.	Autonomy	4.06	.99	.59**	.61**	----									
4.	IT	3.84	1.17	.49**	.57**	.45**	----								
5.	Perf	3.52	.97	.67**	.65**	.62**	.57**	----							
6.	BBuffering	3.34	1.37	.61**	.67**	.59**	.59**	.64**	----						
7.	BSpanning	3.74	1.11	.49**	.63**	.59**	.49**	.52**	.57**	----					
8.	Rewards	3.75	1.17	.63**	.63**	.54**	.51**	.54**	.61**	.51**	----				
9.	Relations	3.44	.98	.16*	.29**	.14*	.17**	.04	.25**	.37**	.18**	----			
10.	SocDes	4.34	.644	.40**	.16**	.40**	.26**	.46**	.29**	.39**	.41**	.03	----		
11.	Age †	1.51	.50	.09	.13*	.01	.18**	.08	.21**	.05	.11	.13*	.00	-----	
12.	Size††	1.84	.84	.17**	.15*	.19**	.05	.27**	.11	.11	.16**	-.25**	.16*	-.09	-----

†Group Age: 1=0-5 years; ††Organization Size: 1=<1,000; 2=1,001-10,000; 3=10,000+ employees

INTERCLASS CORRELATIONS (ICC)

Because our model focuses on the team/group level, we took a number of steps before aggregating individual responses to the team level. As discussed, a number of steps were taken to match the level of measurement to the level of theory discussed through in the development of our survey instrument and selection of constructs.

Additionally, to justify aggregation, we assessed the convergence of responses of team members. I initially created a label and coded 13 groups that had two or more members. The groups were coded with a number of one through thirteen in order to maintain the confidentiality of the groups' identities. The thirteen groups comprised of 72 respondents of the total 252. I filtered the data to analyze one group at a time to evaluate interrater reliability. Interclass Correlations Coefficient (ICC) estimates and their 95% confidence intervals were calculated using SPSS statistical package version 24 based on a 2-way mixed-effects model using a consistency definition. The data were restructured in SPSS in order to measure the extent of agreement of responses among group members across all survey items. Using best practices from "A guideline of selecting and reporting intraclass correlation coefficients for Reliability Research" (Koo & Li, 2016), I recorded the results below in Table 9.

Table 9: Interclass Correlations Coefficients

Group	N	ICC	95% confidence interval		F-value	Significance
			Lower	Upper		
1	5	.671	.161	.447	3.041	.000
2	3	.218	-.283	.544	1.279	.164
3	6	.811	.711	.885	5.293	.000
4	18	.831	.750	.895	5.928	.000
5	6	.657	.474	.792	2.915	.000
6	9	.718	.575	.827	3.541	.000
7	3	.614	.366	.775	2.588	.000
8	5	.610	.395	.765	2.564	.000
9	4	.611	.385	.768	2.569	.000
10	6	.458	.169	.671	1.844	.002
11	5	.377	.034	.624	1.604	.017
12	2	.201	-.454	.561	1.252	.230
13	9	.777	.664	.863	4.483	.000

Only two of the groups did not achieve significance at 95%. These are group 2 and 12, which perhaps not surprisingly, these were the smallest size groups. The largest groups had the highest amount of agreement from members (groups 4, 6, and 13), demonstrating the power of sample sizes. Additionally, group 11 (ICC = 0.377) has an upper bound of 0.624 and lower bound of 0.034, with a p-value that is less than $0 < 0.05$. We can conclude that the ICC is unlikely to be zero (it is significantly different from 0). However, based on Koo and Li's (2016) guidelines, we cannot conclude that this group is significantly strong enough to indicate the group 11 answered the question consistently. For these groups—2, 11, and 12—medians will be used for structural equation modeling (SEM), while means were used for all other constructs. In short, the respondents showed significant agreement, which may also provide increase confidence that single respondents' were likely not far off their own group members.

SEM Aggregated

Structural Equation Modeling (SEM) allows analysis of dependence relationships linking the hypothesized models constructs. The structural model is most useful in representing the interrelationships of variables between constructs (Hair et al., 2010).

Of the 252 respondents, 81 cases belonging to 13 groups were aggregated for structural equation modeling (SEM). The 13 aggregated cases were added to the data set, while 81 individuals belonging to the groups were deleted from the data set for SEM Analysis, for a sample of $n=184$. Before advancing to Structural Equation Modeling (SEM), I imputed factor scores for the 10-factor solution using the data set of aggregated groups.

In evaluating a hypothesized model, it is important to compare its fit to competing models (MacCallum, Wegener, Uchino, & Fabrigar, 1993), as it is “quite difficult to evaluate a single model in isolation without a reference point” (MacCallum, 1998: 22). I proceeded to build two models, and include key fit statistics and explanatory power (R-squared values) of the models. It is essential to examine the overall fit of the model, for if the model does not fit the data acceptably, the overall hypothesis that the model is an accurate representation of the data is rejected, and the coefficients estimated in the model can be biased and are thus meaningless (Judge, Hurst, & Simon, 2009). Fit statistics and squared multiple correlations for the two models are as follows:

Table 10: Model Comparisons

Overall Fit Statistics	M1	M2
SRMR	.009	.006
CFI	1.00	1.00
RMSEA	.00	.00
PClose	.562	.697
CMIN	1.76	2.19
Df	2	3
CMIN/DF	.879	.731
R-Square Rad	.58	.60
R-Square Perf	.55	.59

The first model (M1) contains the independent variables only: relational climate; boundary buffering; boundary spanning; senior management communication/relationships; IT capabilities; and, rewards, in addition to the two dependent variables, performance and radical innovation. The intent of this model is to see whether the relationships are significant for our outcome variables and to note a base model R-square, to see to what extent additions to the model add to its explanatory power. The second model (M2) adds the controls—group age, company size, autonomy, and social desirability. The intent here is to understand the total impact the various controls may have on the direct effects and the R-square of the model.

With regards to RMSEA (Root Mean Square Error of Approximation), all statistics are within acceptable range using a threshold of .05 (Browne & Cudeck, 1993). M1=M2=0. The RMSEAs stand out as small, and therefore excellent. Small values are noteworthy because Kenny, Kaniskan, and McCoach (2015) argue RMSEA for models with small df and low N can have artificially large values of the RMSEA (Kenny et al., 2015). Related to RMSEA and sample size, PCLOSE, p-values of the PCLOSE metric

reflect close-fitting models when the p values are greater than .05 (i.e., not statistically significant), and therefore are close-fitting models. Both models are, therefore, close-fitting, and M2 has the edge over M1. SRMR, defined as the standardized difference between the observed correlation and the predicted correlation, is an absolute measure of fit in which a value of zero indicates perfect fit. A value less than .08 is generally considered a good fit (Hu & Bentler, 1999), making both models ‘good’ along this dimension, and M1 (.009) as best of the model when compared to M2 (.006). Chi-square/df is an old measure of fit, and arguably an important measure for models with small sample sizes (of about 75 to 200 cases, according to Kenny et al. (2015), and with $n=184$ is relevant to my data set. Our models have values under 3, and are, therefore, acceptable. Again, examining table 11 above, M2 shows the better result (.731), vis-à-vis M1 (.879). Moreover, the squared multiple correlations for each structural equation is best in M2, with total variance explained of 60 and 59 percent for Radical Innovation and Performance respectively, vis-à-vis 58 and 55 percent in M1. In conclusion, notwithstanding the model fit acceptability of both models, M2 is the superior model when it comes to model fit statistics while showing a substantially better R-square, and is therefore used for further analyses of direct effects.

RESULTS

Control Variables

I controlled for several relationships that should be taken into account when considering groups’ radical innovation and performance. Those were organizational size, group age, autonomy, and social desirability. Autonomy was only moderately significant ($p<.10$) in its relationship with performance, and not with radical innovation. Social

desirability was positively related to both dependent variables ($p < .01$). In contrast to my expectation that Group Age and Organization Size would be positive and significant, they were not.

Direct Effects

Figure 7, at the end of this section, shows the standardized coefficients for the hypothesized model, and Table 19, immediately below it, provides a side by side comparison of the standardized coefficients for each of the two outcome variables. Results indicate that relational climate only weakly influenced radical innovation ($RC \rightarrow Rad$ of $\beta = .13$, $p < .10$) and **not supporting 1a** hypothesis at $p < .05$; however, it had a positive and significant effect on performance ($RC \rightarrow PF$ of $\beta = .18$, $p < .01$), **supporting Hypothesis 1b**.

As expected, boundary spanning influenced radical innovation ($BS \rightarrow Rad$ of $\beta = .14$, $p < .05$) **supporting hypothesis 2a** and did not drive performance ($BS \rightarrow PF$ of $\beta = .05$), so hypothesis **2b** was **not supported**.

Boundary buffering had positive and significant effects on both radical innovation ($BB \rightarrow Rad$ of $\beta = .22$, $p < .001$) and performance ($BB \rightarrow PF$ of $\beta = .23$, $p < .001$) **supporting hypotheses 3a and 3b**.

Furthermore, boundary buffering was more impactful than boundary spanning in its influence on radical innovation and on performance **supporting hypotheses 4a and 4b**, demonstrating the superior influence of boundary buffering, which is surprisingly missing from the innovation and CE literature over the more studied boundary spanning.

The groups' relationship/communication with senior management did not significantly have an effect on radical innovation ($RL \rightarrow Rad$ of $\beta = .08$). Therefore,

hypothesis 5a was not supported. Sr. management relationship/communication did, however, influenced performance (RL→PF of $\beta = -.21$, $p < .001$), but **in the reverse direction**; therefore, Hypothesis **5b** was **not supported**, but the result **supports** the Hypothesis **6b** designed competing with Hypothesis 5b. In other words, senior management's relationship/communication inversely affects performance, in support of Hypothesis 6b, whereas there is no support for Hypotheses H5b, positing a positive outcome of senior management relationship/communication.

Consistent with **supporting Hypothesis 7**, performance-based rewards for innovativeness positively influenced radical innovation (Rew→Rad of $\beta = .17$, $p < .01$).

Finally, **supporting** Hypotheses **8a** and **8b**, a proactive IT stance positively and significantly influenced radical innovation and performance (IT→Rad of $\beta = .14$ ($p < .01$); IT→PF of $\beta = .27$, $p < .001$).

Table 11: Summary of Results - Standardized Betas of Independent Variables on Dependent Variables

Exogenous Variable	Radical Innovation	Performance
Relational Climate	.13 †	.18**
Boundary Spanning	.14*	.05
Boundary Buffering	.22***	.23***
Sr. Relationships	.08	-.21***
Rewards	.17**	-----
IT Stance	.14*	.27***

*** $p < 0.001$; ** $p < 0.010$; * $p < 0.050$; † $p < 0.10$

Table 12: Direct Effects - Hypotheses Results Table

Hypotheses – Direct Effects	Results
<i>Hypothesis 1a. Group relational climate has a direct positive influence on radical innovation.</i>	Not Supported at $p < .05$; ($\beta = .13$, $p < .10$)
<i>Hypothesis 1b. Group relational climate has a direct positive influence on performance.</i>	Supported ($\beta = .18$, $p < .01$)
<i>Hypothesis 2a. Group boundary spanning has a direct positive influence on radical innovation</i>	Supported ($\beta = .14$, $p < .05$)
<i>Hypothesis 2b. Group boundary spanning has a direct positive influence on performance.</i>	Not Supported ($\beta = .05$)
<i>Hypothesis 3a. Group boundary buffering has a direct positive influence on radical innovation.</i>	Supported ($\beta = .22$, $p < .001$)
<i>Hypothesis 3b. Group boundary buffering has a direct positive influence on performance.</i>	Supported ($\beta = .23$, $p < .001$)
<i>Hypothesis 4a. Top Management relationships/communication (Leaders-Group) positively influence radical innovation.</i>	Not Supported $\beta = .08$
<i>Hypothesis 4b. Top Management relationships/communication (Leaders-Group) positively influence performance.</i>	Not Supported ($\beta = -.21$, $p < .001$)
<i>Hypothesis 5a. Top Management relationships/communication (Leaders-Group) negatively influences radical innovation.</i>	Not Supported $\beta = .08$
<i>Hypothesis 5b. Top Management relationships/communication (Leaders-Group) negatively influence performance.</i>	Supported ($\beta = -.21$, $p < .001$)
<i>Hypothesis 6. Performance Based Reward System (for innovativeness) positively influence radical innovation.</i>	Supported ($\beta = .17$, $p < .01$)
<i>Hypothesis 7a. A Proactive IT stance positively influences radical innovation.</i>	Supported ($\beta = .14$, $p < .01$)
<i>Hypothesis 7b. A Proactive IT stance positively influences performance.</i>	Supported ($\beta = .27$, $p < .001$)

Mediation

Model 3 was created to test for the mediation effects of boundary work variables, per the hypothesized model, on the relationship between relational climate and outcome variables. The fit statistics are included in Table 13.

Table 13: Fit Statistics – Mediation of Boundary work

Overall Fit Statistics	M1	M2 Direct Effects	M3 Mediation 1
SRMR	.009	.006	.022
CFI	1.00	1.00	.98
RMSEA	.00	.00	.043
PClose	.562	.697	.552
CMIN	1.76	2.19	19.21
<i>Df</i>	2	3	13
CMIN/DF	.879	.731	1.34
R-Square Rad	.58	.60	.60
R-Square Perf	.55	.59	.58

I used Preacher and Hayes’s methodology, and I have included the following table summarizing AMOS output of the total, direct and indirect effects with corresponding significance based on two-tail tests, using bootstrapping of 2000 estimates and 95% confidence intervals.

The output from Table 14 indicates whether boundary spanning and boundary buffering mediate the between relational climate and the dependent variables, radical innovation and team performance, and shows that boundary buffering fully mediates the relationship between relational climate and radical innovation and partially mediates the relationship between relational climate and performance, **in support of Hypotheses 8a and 8b**, respectively. Further, boundary spanning fully mediates the relationship between relational climate and radical innovation (**in support of Hypothesis 9a**). Results also indicate that boundary spanning does not mediate relational climate and performance (Hypothesis 9b).

Table 14: Mediation of Boundary Spanning and Boundary Buffering on Relational Climate to RI and Performance

Radical Innovation					Performance			
Variable	Direct	Indirect	Total	Mediation Type	Direct	Indirect	Total	Mediation Type
B Span	.127	.119*	.256*	Full	.173*	.043	.226*	None
B Buff	.143	.183*	.325*	Full	.185*	.179*	.365***	Partial

*** p < 0.001, ** p < 0.010, * p < 0.050, † p < 0.10

Given the diversity of approaches and statistical techniques that currently exist for testing mediation (Mathieu & Taylor, 2006: 27), we opted to use a second method, in line with Baron and Kenny (1986), to see if the results concur. In this case, direct paths were first recorded with the mediators removed from the model, initially with the meditation paths through Boundary Buffering to each of the dependent variables, and then with the mediation path only through Boundary Spanning. The direct effects were then recorded, with each of the mediators added to the model, triangulating results showed previously, and calculating indirect paths using bootstrapping of 2000 estimates and 95% confidence intervals.

Table 15: Direct Effects (with and without Mediators) & Indirect Effects

Relationship	Direct w/o Mediator	Direct w/ Mediator	Indirect	Result
Rel>BB>RI	.148 *	.143+	.183*	Full
Rel>BS>RI	.148 *	.137+	.119*	Full
Rel>BB>PF	.216 **	.185**	.179*	Partial
Rel>BS>PF	.216 **	.182**	.043, NS	None

*** p < 0.001, ** p < 0.010, * p < 0.050, † p < 0.100

Mediation hypotheses results are summarized in Table 16. That boundary work variables are really important (full mediation) suggests that just having top team (i.e., good

relational climate) pursuing its entrepreneurial aims is not enough and has small effects without boundary work.

Table 16: Mediation - Hypotheses Results Table

Hypotheses – Mediation	Results
Hypothesis 8a. Boundary buffering mediates the relationship between relational climate and radical innovation.	Supported, Full mediation
Hypothesis 8b. Boundary buffering mediates the relationship between relational climate and performance.	Supported, Partial
Hypothesis 9a. Boundary spanning mediates the relationship between relational climate and radical innovation.	Supported, Full
Hypothesis 9b. Boundary spanning mediates the relationship between relational climate and performance.	Not supported

SUPPLEMENTAL ANALYSES: PERFORMANCE-BASED REWARDS AND NON-STANDARD REWARDS/INCENTIVES

Three questions that inspired the supplemental analysis of this section are as follows: (1) Do Performance-Based Rewards matter for group performance? (2) To what extent are non-standard (unique) incentives for exploratory groups available? and (3) Which incentives matter to exploratory team members?

Performance-Based Rewards – Is There an Effect on Performance?

The positive and significant influence of Performance-Based Rewards on radical innovation supports my expectation that rewards mechanisms *do* make a difference for radical innovation. While a performance-based reward was not hypothesized to influence performance in this study, I wanted to be sure about its relationship with performance, the other outcome variable.

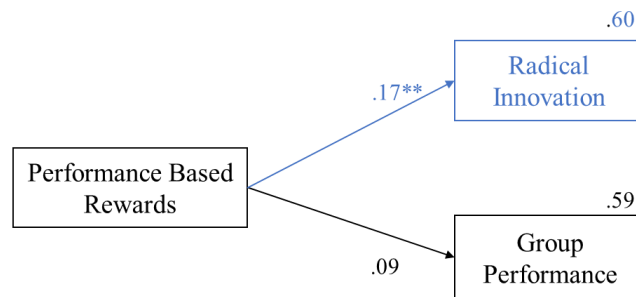
To do so, a path was added to our Model 2 from Rewards to Performance in addition to the path rewards to radical innovation, creating a fourth model (M4). As summarized in Table 17, model fit was excellent.

Table 17: Adding a path Rewards → Performance in M4

Overall Fit Statistics	M2	M4
SRMR	.006	.006
CFI	1.0	1.0
RMSEA	.00	.000
PClose	.697	.998
CMIN	2.19	.007
<i>Df</i>	3	2
CMIN/DF	.731	.996
R-Square Rad	.60	.60
R-Square Perf	.59	.59

The full SEM model is depicted in Appendix H. Zooming in on this variable, the results are illustrated in Figure 9, with standardized estimates showing that performance-based rewards was insignificant, demonstrating that an insignificant relationship of performance-based rewards with performance.

Figure 9: Paths Rewards to Performance (and RI)



*** $p < 0.001$; ** $p < 0.010$; * $p < 0.050$; † $p < 0.100$

To What Extent Are Non-Standard (Unique) Incentives Available for Teams

Exploring Non-Standard Directions?

The survey included the question “the compensation model is basically the same for me/my Group as it is for employees working in the established company/organization” with responses of “yes”/“no.” This study finds that 77% of the innovation group managers are compensated on a similar basis as other employees not working on entrepreneurial initiatives. This finding mirrors the regrettable findings of prior research on radical innovation finding that reward systems customized to encourage radical innovation in entrepreneurial units to be rare, and as stated by O’Connor and McDermott (2004) that radical innovation teams were compensated no differently from anyone else in the firm, which is contrary to the position that non-standard compensation systems are needed, a case widely asserted in the literature (e.g., Hisrich & Kearney, 2011).

Non-Standard Incentives: Which incentives Matter to Exploratory Team Members?

Which Are Available?

Given that the construct, *performance-based rewards*, does not offer specific actions that can be pursued to incentivize the entrepreneurs within established organizations, I included questions to identify incentives of interest to respondents (per Appendix F), as discussed in my literature review. Respondents were asked to rank the importance of eleven incentives on a scale of 1 through 5, and whether it was available (yes/no).

Table 18: Ranking of Incentives and Availability

Importance	Availability
Access to talented people in total company	Access to talented people in total company
Total company Reputation	Total company Reputation
Career progression support	Access to Financial Resources
Schedule Flexibility	Potential job opptys in total co/other units
Customized incentives based on group's unique circumstances	Schedule Flexibility
Access to Financial Resources	Location Flexibility
Protection from downside risk in the event venture changes.	Career progression support
Customized incentives based on group's LT success	Protection from downside risk in the event venture changes.
Location Flexibility	Customized incentives based on group's LT success
Potential job opptys in total co/other units	Equity in group's venture
Equity in group's venture	Customized incentives based group unique circumstances

Note: Equity metrics likely reflect the relatively small percentage (33%) of participants employed by publicly traded organizations included in the sample, and should not be interpreted as unimportant.

Most important. The top two most desired attributes relate to the total company.

These are company-related: (1) Total Company Reputation, and (2) Access to talented people in the company. The next three most desired attributes suggest ways that the company can uniquely support exploratory team players. They are (3) Career progression support, (4) Schedule Flexibility, and (5) Customized incentives based on the group's unique circumstances.

Available vs. most important. The largest gaps when it comes to those attributes that are important to respondents and those that are available are, *career progression support* and *customized incentives based on group's unique circumstances*, thereby providing the most impactful opportunity to close the largest gaps

As a penultimate step, I attempted to reduce the number of incentive variables from the original set of eleven incentive-oriented attributes. To do so, I performed Principal Component Factoring in SPSS. Principal components analysis is a method of data reduction. As seen in Appendix I, in order to reach 95% of the variance, PCA will need to extract ten components. Having started with an original set consisting of 11

features, theirs is negligible reduction. Thus, PCA will not help in terms of dimensionality reduction. In retrospect, this result may be an important finding in its own right, supporting the nuanced character of the i-deal which by definition—voluntary, personalized agreements of a nonstandard nature negotiated between individual employers and their employees regarding terms that benefit each party (Rosen et al., 2013: 710)—may not lend itself to an easily prescribed set of attributes. Rather, this lack of pattern is in keeping with the discretionary component of i-deals where room is given to supervisors and employees to craft their own incentives, a feature which is supported Organizational Support Theory (OST) suggesting that the discretionary nature of favorable organizational treatment is what leads employees to experience the positive outcomes of organizational support (Rosen et al., 2013).

Open-ended responses. An optional question asked respondents to add comments about “overall comments about rewards/incentives to bring out your best.” Forty-five respondents replied, citing 55 incentive attributes. I categorized the responses into the following five categories or themes: life/work balance, rewards/recognition, individual and company performance-based incentives, miscellaneous, and comments about the absence of non-standard incentives. Appendix J lists each of the responses by theme.

Mismatched incentives to risk. Standard incentives for non-standard groups. Two responses from innovation groups within multi-unit corporations reflect the overall assumption of group members’ disproportionate downside risk vis-à-vis the upside rewards.

- The team right now operates at "full throttle" all the time because of our commitment to our project and our fellow team members, but the feeling of possessing all the downside risk of a start-up (failure and team termination)

without the upside is always present and must be managed to maintain team morale. This can create turnover issues on small teams where single team member departures have an outsized impact on team performance compared to larger organizations. (innovation unit, diversified MNC)

- There is a lot of risk in R&D with the fear of losing my job if the program does not succeed. There are a lot of great Engineers that would benefit from the "incentive" of knowing that they can take this risk but still have ABC Co provide us a job opportunity if the business is not successful. (innovation unit, diversified MNC)

Summary of comments. An example of the performance-based *monetary/performance-based rewards* includes: “I believe that better visibility into the value end-users of products we design and develop will motivate higher quality thinking, better work environment, and more satisfied employees. I also think direct financial incentives in the long-term success of an innovation initiative would help group team members better connect to the work, make better decisions (if measured appropriately), and keep each other accountable” (Innovation, Design & Development leadership team).

Examples of the *life/work* balance category include: “In regards to motivation, quality of life and flexibility with family is more important than excessive compensation (generic pharma breakthrough team); and, “Work from home would increase focus” and, “Work from home would increase focus due to less walk up disturbance” (wholesaler-distributor, special projects).

Examples of *non-monetary incentives* include “caregiver incentive program” and “service credit (market expansion team, university).

Finally, examples of respondents voicing the presence of standard incentives or lack of non-standard incentives/rewards, include: “Currently no special incentives for

group. Will change for 2019” (head of company and team charged with transforming from paper to media company).

In summary, the most common response related to monetary incentives. In fairness, while I was usually able to ascertain that the incentives were in place (“start-up group has incentives based on new business”) or not (i.e., ‘financial incentives in LT success would help), it was not always clear, but the popularity of this theme is nonetheless noteworthy. The second most popular theme had to do with the lack of customized rewards and the prevalence of ‘standard’ rewards. This is no surprise given the finding earlier that 77% of breakthrough groups in my data set are compensated the same as all other employees. However, it seems to bring out a very conscious, ‘top-of-mind’ awareness of the omission of non-standard rewards and incentives among employees and should serve as a wake-up call to senior leaders.

After monetary-based incentives, work/life balance theme contained the most replies for desired incentives/rewards. All entries arguably relate directly or indirectly to scheduling flexibility. Recalling that ‘scheduling flexibility’ received high scores—fourth highest—on the scaled questions, shows the need to pay attention to scheduling flexibility in the context of work-life balance context. Furthermore, given that the survey was administered prior to the current pandemic and related world-wide lockdowns, likely means that importance of this attribute may have been understated.

POST HOC ANALYSIS: MEDIATION

Mediation: Could Radical Innovation Play a Mediating Role?

The role of radical innovation as an outcome variable in this study was superbly suited to the context of this study and justified in the selection of my constructs. Upon

reflection, radical innovation is a factor that could conceivably link antecedents to performance. Although my literature review yielded that no one has tested the mediation of the relationship of antecedents in and performance, the link between innovation and performance is well established in the literature (Han, Kim, & Srivastava, 1998), which suggests that innovation is positively related to performance outcomes of this analysis—including subjective performance. Vincent et al. (2004) went further and demonstrated, based on synthesizing 23 years of innovation research findings from economic, strategy, and marketing literature through meta-analysis, that innovation is a partial mediator of the relationships between certain organizational and environmental antecedents and firm performance. As such, I find it worthwhile to examine whether paths from antecedents to Performance could be mediated by Radical Innovation in my sample. Along these lines, could radical innovation be a mechanism through which the antecedents work to influence performance, in which: 8a) Radical Innovation mediates the effect of Relational Climate on Performance; 8b) Radical Innovation mediates the effect of Boundary Buffering on Performance; and 8c) Radical Innovation mediates the effect of Boundary Spanning on Performance.

Mediation Results

To test this, a fifth model (M5) was added to my models. M5 seeks to understand mediation by adding a path from radical innovation to performance. As seen in Table 19, good model fit was established.

Table 19: Fit Statistics – Mediation 2 (M4)

Overall Fit	M1	M2 Direct Effects	M3 Mediation 1	M4 (M2+Rew)	M5 Mediation 2
SRMR	.009	.006	.022	.006	.006
CFI	1.00	1.0	.98	1.0	1.0
RMSEA	.00	.00	.043	.000	.020
PClose	.562	.697	.552	.998	.492
CMIN	1.76	2.19	19.21	.007	2.15
<i>Df</i>	2	3	13	2	2
CMIN/DF	.879	.731	1.34	.996	1.07
R-Square Rad	.58	.60	.60	.60	.60
R-Square Perf	.55	.59	.58	.59	.57

This model built shows a weak, insignificant direct effect from RAD to PERF, which means that mediation of the independent variable through radical innovation to performance is not significant. To illustrate, I calculated the indirect paths using bootstrapping of 2000 estimates and 95% confidence intervals.

Table 20: Mediation Paths

Indirect Paths to Performance	Estimate	Lower	Upper	p-value
RC x RAD	.002	-.018	.037	.687
BB x RAD	.004	-.034	.048	.745
BS x RAD	.002	-.019	.039	.709
REL x RAD	.001	-.011	.023	.582
IT x RAD	.002	-.022	.037	.791
REW x RAD	.003	-.024	.054	.737

The paths are insignificant, as seen below, thereby confirming the lack of mediation and lack of support.

Table 21: Supplemental Mediation Hypotheses Results

Hypotheses – Mediation	Results
<i>Hypothesis 8a. Radical Innovation mediates the effect of Relational Climate on Performance.</i>	Not Supported
<i>Hypothesis 8b. Radical Innovation mediates the effect of Boundary Buffering on Performance.</i>	Not Supported
<i>Hypothesis 8c. Radical Innovation mediates the effect of Boundary Spanning on Performance.</i>	Not Supported

DISCUSSION, IMPLICATIONS, AND LIMITATIONS

Almost thirty years have passed since Kuratko began developing antecedents to corporate entrepreneurship, which later evolved into five antecedents (Goodale et al., 2011). This set of variables resulted in an instrument that has benefitted practitioners and researchers of corporate entrepreneurship. However, dramatic changes have advanced and changed innovation and entrepreneurship inside and outside of organizations, as discussed in this paper. The coronavirus era magnifies the impact of sudden blows to existing business operations and business models, in addition to the more gradual influences of hyper-competition on existing organizations, raising the importance of nurturing and developing new directions.

Taking a fresh practice-based view of the problems associated with implementing successful non-incremental, entrepreneurial initiatives in established, large organizations, this study expanded on a tried and true measure of corporate entrepreneurship (performance-based incentives) while proposing an updated, multi-layered model of four additional variables. The study is unique in incorporating relational climate and boundary work (spanning and buffering), adding IT capabilities into the model, and combining constructs from organizational behavior (OB), innovation, and corporate entrepreneurship

(CE) under a single theoretical framework founded on dynamic capabilities, and ambidexterity. As such, it has the potential to start a new stream of research that builds on important drivers and levels of the organization that influence the successful management of entrepreneurial groups' lofty pursuits, and methodologically contributes by capturing the often-missed and distinct group/team unit of analysis and perspective vis-à-vis the larger organization. A discussion of specific drivers follows.

Relational Climate

In my data set, we saw that groups high in relational climate—their interpersonal interaction and relationship quality—are higher performing. My hope is to feed research on interactions as an impetus for ongoing corporate entrepreneurship, as I will discuss shortly. For leaders, this means placing emphasis on group unity, resonance, and communication among members of these groups, and making space for interactions and nurturing and keeping tabs on the group dynamics and morale.

Boundary Buffering and Spanning

This study also sought to demonstrate the relevance of team level boundary buffering—the lesser-known dimension of boundary work—in the context of corporate entrepreneurship. By definition, boundary buffering refers to leadership's minimization of the group's external interference. The positive, very strong and nearly equal influences on radical innovation and performance seems to indicate that buffering, an action that that is undertaken in response to or in anticipation of disruptive forces within an environment (Faraj & Yan, 2009: 606) is a not only a powerful influence, but also actionable for leaders seeking to pursue corporate entrepreneurship.

The importance of boundary work was discussed throughout this research, and directly relates back to the ‘problem of practice’ that motivated this research journey. Specifically, the problem centered around entrepreneurial teams/units that do not reach their potential when they run up against established ways of doing things within the larger organization(s) in which the group is embedded. The findings from mediation analyses reinforce the importance of not only building the team relational climate but also going further to orchestrate spanning and buffering of the team in relation to the larger organization.

Together with boundary spanning, I hope the study contributes to shifting from a management emphasis on the formal structure of innovation units to one focused on ways to effectively provide entrepreneurial teams not only with access to resources (as defined by boundary spanning in this study) but also to protect the unit from interference demands and interference.

In short, for researchers, I argue for group boundary buffering and spanning as a more relevant measure vis-à-vis formal organization structure, in light of arbitrary structuring decisions, and amid increasingly blurring and porous organizational boundaries. The boundary work variables represent actions that could be implemented in teams and organizations without regard to physical location, in virtual teams that are here to last long after the current pandemic. Furthermore, for practitioners, the results of this study can help practitioners see the importance of ensuring buffering and spanning innovation groups. Given the trade-offs in time and effort required to manage the competing demands of a team’s internal and external processes (Choi, 2002; Faraj & Yan,

2009), organization and team leaders could monitor its activities in both realms, ensuring that buffering is not overlooked when necessary when Radical Innovation is the goal.

Sr. Management-Group Relationships

The Sr. Manager Relationships construct, which probes frequency of communication with two senior managers, is really a construct of communication frequency, which was used the study of radical innovation (O'Connor & DeMartino, 2006) as a proxy for relationship quality. The strong inverse relationship in the context of this study raises questions about the relationship *quality* aspect the measure and begs the question as to possible explanations for the negative direction of the coefficient.

Our findings largely make a case for the delicate character of communication with/from senior executives. It is plausible that frequent communication with senior management is perceived as a form of interference as with any other external noise. This explanation seems consistent with Choi's observation of well-developed teams in which the author offers the view that 'directive interventions by leaders could often be redundant or even distracting for members,' supporting the team literature that suggests the cons of overmanaging groups and undermining its autonomy (Druskat & Kayes, 1999). The opportunity to dig deeper and identify the kinds of relationships (i.e., relational climate) and types of communication represent fertile ground for discovery and impact. As discussed in the literature, top-down flows that work for exploitation units could be detrimental to radical innovation teams. Looking more closely at the specific type of communication and relationships could prove extremely fruitful.

One possible explanation could stem from a positive perception from group members that the infrequency of communication is just what the group needs—the

‘space’ to get its job done. Indeed, infrequent communication could signal a sign of confidence from the upper echelon. Along this vein, it might be helpful to remember that the innovative unit runs contrary to the established parts of the company, and it is not uncommon for the unit to be inconsistent with the ‘vision’ of the rest of the organization. If successful, it’s quite possible that the innovative unit will re-shape the organization’s vision, making reconciliation of dissonant visions and pursuits something that is difficult, if not unnecessary, to communicate.

It could be that senior management is simply not in sync with the timing of the payoff associated with the entrepreneurial effort. I will call this mismatch a lack of strategic synchronicity. After all, the mortality rates of young, unproven venture units are sufficiently high (Dushnitsky, 2012) and there is the widespread perception of corporate venturing (CV) units as highly vulnerable entities that are frequently terminated before they are able to demonstrate returns (Burgelman & Välikangas, 2005; Fast & Pratt, 1981; Gaba, 2007). As a point of reference, research into the Venture Capital (VC) industry, on which corporate venturing unit practices have been built, shows that VC funds require several or eight years before they are able to show a yield on their start-up investments (Gompers & Lerner, 2001a, 2001b). By comparison, the cycle of leadership change in large organizations is shorter, with the average tenure of the CEO of a large company lasting only six years (Kaplan & Minton, 2006). It is conceivable, therefore, that executives in large organizations will make negative judgments regarding the fate of their entrepreneurial initiatives before groups have had time to show whether they have been successful or not (Burgelman & Välikangas, 2005; Dushnitsky, 2006).

Less likely, in my opinion, although a possibility, is a situation in which senior management communicates and relates with the initiative often and effectively, but that other employees from the core or other units do not support the initiative. They could possibly resent it. Respondents could feel like communication from senior managers feel like communication and support from leaders actually makes their life worse. Questions in future studies to both team members and non-team members about senior management intervention and attitudes could help determine whether this explanation that what I refer to ‘malignant envy’ is plausible.

Incentives

Evidence tilted the score in favor of the positive influence of performance-based incentives on radical innovation, supporting my contention that prior studies, not clearly differentiating between innovation and non-innovation activities and members, may have contributed to the mixed findings. Specific company-wide attributes are important to group members (i.e., company reputation, talented people); however, so are career progression support, schedule flexibility, and customized group incentives. The work/life balance theme was also highlighted, all providing much-needed specifics to this topic. Equity, an attribute cited by 129 respondents as important is very likely to be under-reported as equity is only available to approximately an equal number of respondents who work for public companies. That is, the availability of equity likely reflects the relatively small percentage (33%) of participants employed by publicly traded organizations included in the sample, and should not be interpreted as unimportant. A term such as ‘royalty’ might make more sense than ‘equity’ in exploring and assessing long term rewards relevant and understood by a wider audience.

Proactive IT Stance

This dimension of IT capability proved to influence both innovation and performance (the strongest influence of the study), supporting the notion that IT capability generally impacts radical innovation and related competencies are a critical part of the environment where radical innovation efforts unfold. This underscores the advice for practitioners to continuously morph capabilities in changing environments which can enable continual learning and renewal and thus avoid falling into lock-in rigidity (Choi, 2002; Haeckel, 1999).

IMPLICATIONS AND A MODEL FOR PRACTITIONERS

Understanding group-level phenomena was clearly the emphasis in this study. Organizational-level studies of corporate entrepreneurship are abundant and are out of my study's main scope, although require mentioning in order to glean real-life phenomena and implications that cross-levels. Specifically, certain researchers have discussed the need for a proactive, constant push for CE (i.e., Ireland et al., 2009). This is congruent with my preceding discussion as well as my advice for practitioners. Also consistent with my observations are studies that assert that CE strategies are a matter of degree, with stability on one end and chaos on the other extreme, recognizing the shortcoming of extreme chaos. In spite of these similarities, however, there are distinctions in our perspectives. Ireland et al. (2009) consider CE a distinct organizational strategy of "three foundational elements of an entrepreneurial strategic vision, a pro-entrepreneurship organizational architecture, and entrepreneurial processes and behavior as exhibited throughout the organization" (p. 38). That model asserts the need for CE to

run deep in the organization, and CE is a unique and identifiable strategy that stems from a top-level directed strategic vision that is organization-wide.

Our perspective, based on group-level initiatives pursuing breakthroughs within the established organization, is one based on groups and units that are small in comparison to the core business(es), usually emerging subsets of the organization. As such, CE as a company-wide model or strategy may be a long way off in the context of sitting side by side with exploitative units. My perspective then for practitioners, is different from company-wide approaches, as asserted by Ireland et al. (2009). While it could prove to be complementary to theirs', mine is rooted instead in groups positioned in smaller pockets of the organization, in subgroups like the cross-functional groups I studied as well as those more akin to Zahra's (2015) entrepreneurial hubs.

The term radical innovation may conjure notions of radical if not 'guerilla' implementation. However, the art and science of effectively managing bold, breakthrough innovations is one of finesse, balance, and learning. It warns of the dangers of 'biting off more than could be chewed.' It is hard for me to imagine implementation radical new initiative(s) across established, especially multi-unit organizations all at once without allowing for learning to be absorbed and proper customizations to be made in different units and according to the variety of contexts of the organizational units.

Company-wide, senior-level understanding and buy-in, nevertheless, could play a formidable role in its effective management and success across an organization. For those managers and senior leader practitioners who are contemplating bringing out the best in their groups' initiatives, in the context addressed, we recommend consideration of the following:

1. Taking a view that cross-functional group initiatives, and the individuals serving on them, can serve as pathways that create and convert knowledge and new strategies could have both short- and long-term benefits. Borrowing from Zahra, they create and exploit ‘this heterogeneity to create distinctive products, innovative business models, and well-protected and lucrative market spaces’ (Zahra, 2015: 730). Strategy, of course, is within the domain of senior leadership, and orchestrating the interactions and relationships that carry and convert knowledge within and beyond groups as to channel ‘contagion’ of learning then become topics of conversation at the highest levels. They can determine, for example, when it could be important to direct a team from an internal focus to a more extra-group approach so as to spread innovation initiatives to the next phase, initiative, or unit.
2. Providing the group with the ‘space’ and leader to provide an environment where interactions and relationships can create the kind of relational climate discussed herein which can drive and motivate change within and beyond the group. Rotating members in and out of new initiatives as carriers of valuable knowledge.
3. Communication with the group (and employees) is delicate. Topics taken for granted at an ‘organizational’ level addressing the very company vision or mission may not resonate with the group pursuing breakthroughs, and vice versa. My own study’s challenges in recruiting the appropriate sample and challenges associate with differentiating between group and organization levels in interpersonal and written communications are indicative of the difficulty that senior leaders have in communicating to disparate groups going in different directions—some fulfilling missions that may not only be on the periphery but may move in a contrary direction to the ‘core’ business.’
4. There is no one organizational form that gets plugged in across units or organizations. Cross-functional teams, units, ‘entrepreneurial hubs’ (whatever the desired form based on the context) can stand-alone as separate financial/legal entities, or not, are all possibilities through the group’s evolution. They could be virtual or remote groups, or on-ground or a hybrid. We would argue in favor of digging deeper into what *operationalizing* appropriate linkages to other resources and especially proper buffering, with the ongoing balancing of spanning with buffering.
5. At the organizational level, we would also argue that the extent of integration among different units could take on different forms, as we hoped to convey in our metaphor of various wheels of different sizes and rotations in orbit, and as we illustrated with examples of the creation of entirely new and unique business models. We hope we were clear that the goal of integration is not to have the team and established org become ‘one,’ even though the touting of ‘one team-one company-one vision’ may sound attractive. It could be the case that unique business models, such as international entrepreneurial units as an

extreme example, remain distinct, rotating in opposing directions, to build on the wheels-in-orbit metaphor.

6. The planning and discussion of IT capabilities in a proactive way ('a seat at the innovation planning table') could be instrumental to the initiative's success. Discussion of the role of IT platforms both for the CE initiatives/units and their relationship with other units naturally merits senior managements' attention.
7. Constant balancing of short and long term, variable, and fixed incentives that do not compete with other business units but that are customized to match the goals and objectives of the group, as well as larger organization(s) in which the group is embedded or based upon the degree of independence. Consider and acknowledge intrinsic rewards (i.e., job paths, flexible schedules), and other incentives (i.e., work-life oriented), including idiosyncratic deals, when possible.

Limitations

A limitation is that my measurement instrument relies heavily on self-reported data, with predictor and outcome variables reported by a single source. The journey underscored several difficulties of capturing group (team) level data from diverse organizational settings and from multiple sources in large enough samples to attain statistical power needed for quantitative analysis. There was one team taken from each organization, with the exception of one organization that provided three teams. The small sample size per firm made a fixed effects model of analysis statistically impractical; however, a future study could explore a fixed effects model that remove any potential pernicious effect of omitted variable bias. Also, although I did take several steps and procedures were taken in order to differentiate the group from the rest of the organization. It would be optimal for future studies' samples to be larger and for all groups to include multiple members. Additionally, the data was collected via two different sources, and as mentioned in my test of heterogeneity, one factor, Information Technology items, had means that were significantly different across groups. Finally,

outcome variables would ideally be measured from a knowledgeable, third source. It was not feasible to collect conventional performance metrics, as that would preclude an important subset of non-autonomous, non-revenue, and profit-generating groups and teams from the sample. Performance was subjective performance. However, in the meta-analysis (Vincent et al., 2004) of innovation and performance, results show that ‘innovation is positively related to all three performance outcomes’ (Vincent et al., 2004: 7), including efficiency, subjective performance, and financial performance in order of significance.

Future Studies

Of course, relational climate is not the sole team attribute that matters at the group level. There are many other intra-team attributes that could determine performance outcomes, and should be considered for future studies. To illustrate, team faultlines, shared mental models, geographic dispersion and stability, represent four different group attributes that can be considered.

For example, *team dispersion* is notably relevant in this era in which remote among geographically dispersed team members is ever growing. Introducing challenges to communication that may negatively influence team performance (Eisenberg et al., 2019: 348), a multi-group analysis could dispersed versus non-dispersed teams performance and radical innovation efforts. Alternatively, as modeled in this study, the boundary work variables could serve as mediators, but in this case, mediating between team dispersion and radical innovation. Team dispersion could also serve as a moderating role of relational climate and outcomes.

Second, it has been asserted that ‘diversity is an inherent characteristic of modern organizations’ (Rico, Molleman, Sánchez-Manzanares, & Van der Vegt, 2007: 127), prompting research are the simultaneous effects of *diversity faultlines* and team task characteristics. The faultline model shows how differences between team members affect team functioning and effectiveness and finds that when diversity attributes correlate highly, it becomes more likely that a team will split up into subgroups that might have difficulties cooperating or even start competing against each other (Rico et al., 2007: 127). It could be fruitful to examine the interaction of senior management communication and faultlines on innovation outcomes.

A third example of alternative team attributes stems from *shared team mental models*, which can also have an effect on team performance. Measures of mental models include knowledge content and knowledge structure which meta-analyses (Dechurch & Mesmer-Magnus, 2010) show are predictive of team performance, and could be used in future studies to reflect intra-team characteristics.

A final example of an attribute, *team stability*, introduces a dimension that could be quite revealing amid the context, as discussed throughout this research, in which radical innovation team members are assuming increased risk without proportional reward. Rather than treating turnover as an exogenous variable (e.g., Cornelli, Simintzi, & Vig, 2016), I would suggest that turnover as an endogenous variable could be even more meaningful.

As further fuel for future studies, the connection of different types of group dynamics to organizational learning could be quite fruitful. It could build on the interactions identified by Zahra (based on interactive expertise and transactive memory)

while adding aspects of the relational climate construct relating to empathy, support, and relational energy among members.

As for additional topics for analysis, the dependent variable, radical innovation, serves quite well with a variety of industries and kinds of demonstrable change I sought to study. The group performance measure filled a void in capturing early performance when metrics for mature organizations (i.e., sales, number of employees, and firm profitability) widely used indiscriminately in studies are not suitable. However, it is also conceivable that different strategies (i.e., international expansion), sectors (i.e., non-profit) or industries, be suited to more specific types of performance measures, raising the need for more nuanced dependent variables in the study of breakthrough initiatives within established organizations.

The finding that senior management communication had a negative influence on group performance begs to delve into a more nuanced understanding of the types of communication that influence radical innovation. Mixed methods studies of Qual → Quan designs could explore types of communication styles and relationships between senior leaders and groups, and then test them by examining the moderating or mediating effects of senior management communications on radical innovation. Through longitudinal studies, it would be fruitful to examine temporal differences in boundary buffering and spanning, and senior-level relationships over time by using group age as a moderator.

Boundary work mechanisms of buffering and spanning could benefit from further work. We established the discriminant validity of the two measures in a CE/RI context. Having adopted the construct from the team literature, we suggest building more content

around the actions that constitute team boundary spanning and buffering in the context of breakthrough teams' pursuits.

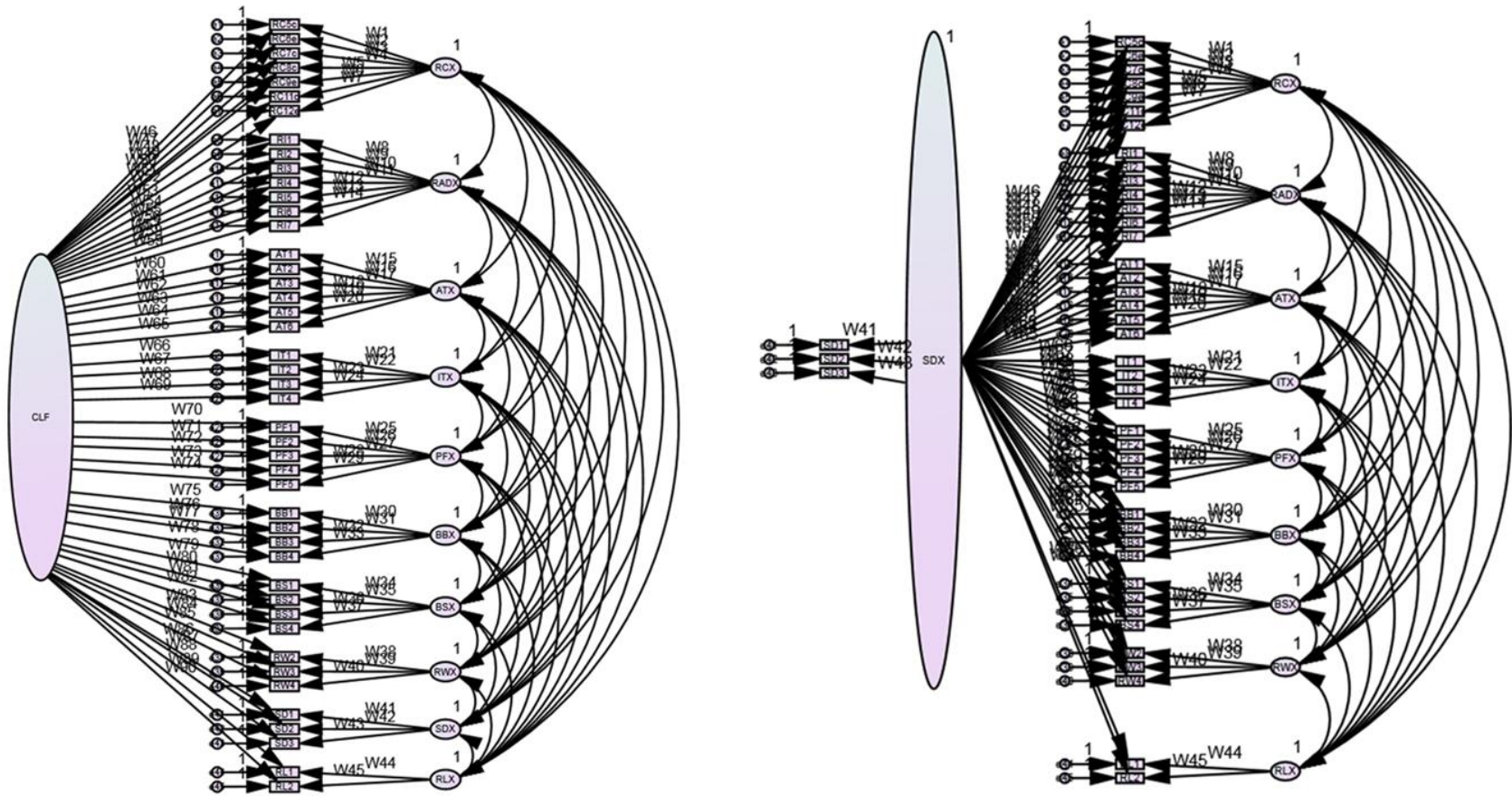
For practitioners and scholars, I hope that my use of knowledge from several fields, namely the CE and innovation management, helps widen the knowledge pool to draw from for advancement of the practitioner-scholar work. Furthermore, that those noted obstacles in this research—are reminders that tried and true traditions of a domain do not always neatly fit the problems of practice. For example, we challenged assumptions associated with applying innovation management concepts and constructs to some aspects of this thesis (i.e., such as the interaction term mixing exploration and exploitation in the common use of dependent variables, and the product development emphasis, in both innovation and CE traditions) which did not always fit the multi-functional, new business models of the breakthroughs innovation this context envisioned. Likewise, applying 'tenets' of CE (i.e., performance rewards) were questioned, and may provide terrain to be explored in future studies.

Appendix A: Pattern Matrix

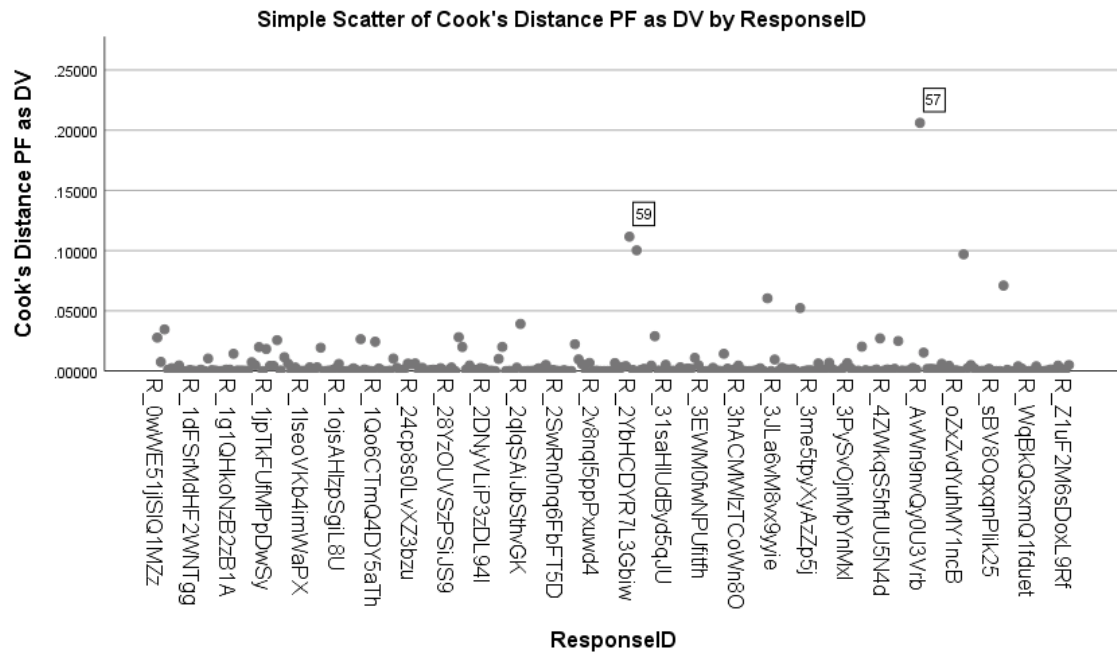
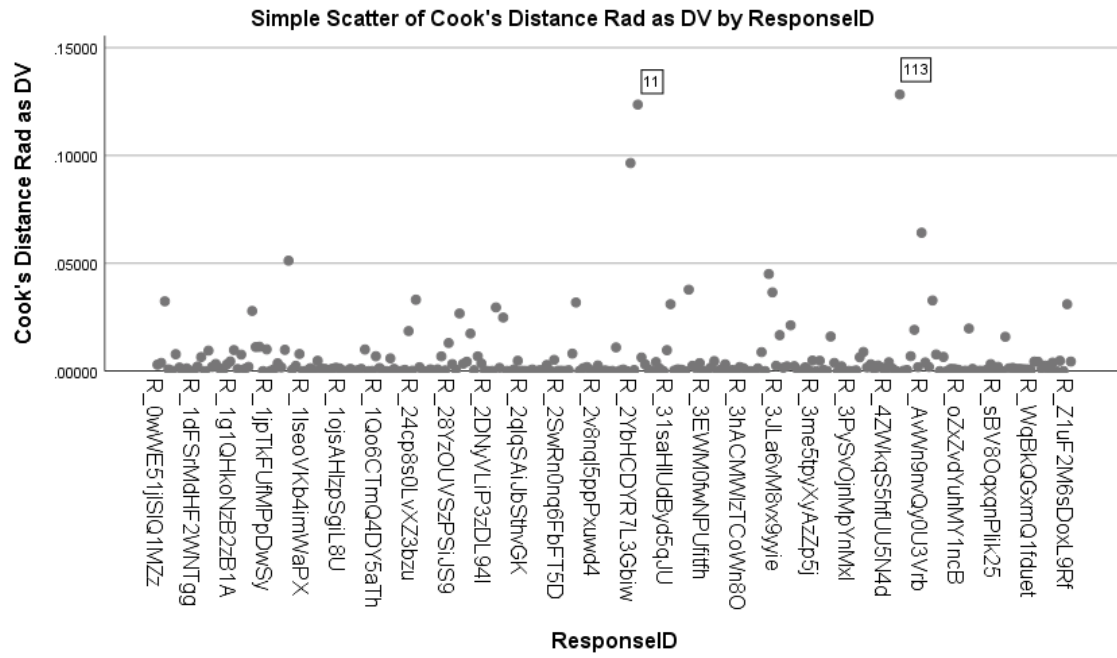
(Principal Axis Factoring, Promax); 10 factor Model 45 items PAF Promax

	RC	Rad	ATY	IT	PF	BB	BS	REW	REL	SocD
Cronbach's Alpha	0.93	0.89	0.90	0.88	0.91	0.83	0.82	0.86	0.78	0.629
RC5c	.85									
RC6e	.81									
RC7c	.90									
RC8c	.64									
RC9e	.88									
RC11c	.77									
RC12e	.69									
RI1		.77								
RI2		.61								
RI3		.74								
RI4		.78								
RI5		.59								
RI6		.63								
RI7		.83								
AT1			.75							
AT2			.69							
AT3			.85							
AT4			.92							
AT5			.68							
AT6			.79							
IT1				.74						
IT2				.88						
IT3				.78						
IT4				.85						
PF1					.86					
PF2					.84					
PF3					.62					
PF4					.73					
PF5					.73					
BB1						.58				
BB2						.78				
BB3						.66				
BB4						.69				
BS1							.76			
BS2							.67			
BS3							.62			
BS4							.69			
RW2								.56		
RW3								.71		
RW4								.80		
RL1									.84	
RL2									.76	
SD1										.55
SD2										.73
SD3										.42

Appendix B: Common Method Bias Tests Using Unmeasured Latent Variable and Social Desirability Marker



Appendix C: Simple Scatter of Cook's Distance



Appendix D: Summary of Measures

Performance (2 scales):

Source: Hackman as adapted by Edmondson (1999)

1. This group (team) does superb work.
2. This group (team) meets or exceeds its (customers') expectations.
3. Critical quality errors occur frequently in this group's (team's) work. (DELETED)
4. This team/organization keeps getting better and better.

Source: Gibson and Birkinshaw (2004)

5. This group (business unit) is achieving its full potential
6. People at my level are satisfied with the level of group's (business unit's) performance.

Exploratory Innovation:

Source: Jansen et al. (2006); Cronbach's alpha of .86; exploratory innovation of the unit.

1. Our unit accepts demands that go beyond existing products and services.
2. We invent new products and services.
3. We experiment with new products and services in our local market.
4. We commercialize products and services that are completely new to our unit.
5. We frequently utilize new opportunities in new markets.
6. Our unit regularly uses new distribution channels.
7. Our unit regularly searches for and approaches new clients in new markets.

Boundary Spanning:

Source: Faraj and Yan (2009); Cronbach's alpha of .81.

1. To what extent does the group encourage its members to try to influence important actors elsewhere in and/or beyond the group on behalf of the group and its work?
2. To what extent does the group value group members for making use of their relationships with others on behalf of the team?
3. To what extent does the group value group members for making use of their relationships with others on behalf of the group?
4. To what extent does the work of the group depend upon information and resources actively solicited by group members, that is, information and resources beyond what comes through official channels?

Boundary Buffering:

Source: Faraj and Yan (2009); Cronbach's alpha of .82.

1. To what extent are outside pressures deflected or absorbed so that the group can work free of interference?

2. To what extent are outsiders prevented from “overloading” the group with either too much information or too many requests?
3. To what extent does the group or group leader help members manage demands placed on them by other organizational units?
4. To what extent do you feel that group members work in a well-buffered or protected environment.

Relational Climate:

Source: Rochford and Boyatzis (2016); Cronbach’s alpha of .96

1. My group’s daily work aligns with our vision. (removed)
2. My group’s purpose is clear. (removed)
3. Members of my group have a shared purpose.(removed)
4. My group’s actions are guided by a shared vision.(removed)
5. Members of my group have similar visions of the group’s future.
6. Members of my group are empathetic toward each other.
7. People in my group notice when others are in need.
8. Members of my group care about each other’s well-being.
9. When someone in my group is in need, my group takes action to assist them.
10. The relationships in my group are a source of energy. (removed)
11. The atmosphere in my group is vibrant.
12. Interactions in my group are lively.

Performance Based Reward System (for innovativeness):

Sources: Souder (1981); Fry (1987); Cissell (1987); Sykes and Block (1989); Kuratko et al. (2005), as used and cited by Alpan et al. (2010); Cronbach’s alpha of .92.

1. The rewards that group members received or will receive are dependent on their work. (removed)
2. Group members will be appreciated by their managers if they perform very well.
3. Group members from every level (of my unit) will be rewarded, if they innovate.
4. Group members with innovative and successful projects will be highly rewarded.

Relationship/(frequency of communication) of group with senior executives:

Source: Hill and Birkinshaw (2014); Cronbach’s alpha of .67.

Indicate the frequency with which you, or your group, communicates with the following individuals or other groups.

1. Senior executives in the corporate parent your group reports (directly) to.
2. Other senior executives of corporate, other groups, or units.

Proactive IT Stance: (A dimension of a IT Capability construct):

Source: Lu and Ramamurthy (2018); reliability of .91.

Relative to other firms in your industry, please evaluate your capability in acquiring, assimilating, transforming, and exploiting IT knowledge in the following areas on a 1 to 5 scale (1 = strongly disagree, 5 = strongly agree).

1. We constantly keep current with new information technology innovations.
2. We are capable of and continue to experiment with new IT as necessary.
3. We have a climate that is supportive of trying out new ways of using IT.
4. We constantly seek new ways to enhance effectiveness of IT use.

Team Autonomy:

Source: Langfred (2005); Cronbach's alpha of .91. (Removed 2 items of lowest Cronbach's alphas).

1. The group can decide when to do particular activities.
2. The group is free to choose the method(s) to use in carrying out work.
3. The group has control over the sequencing of team activities.
4. The group has control over the scheduling of group-work.
5. The group is able to decide its objectives.

Social Desirability:

Source: Crowne and Marlowe (1960).

1. I'm always willing to admit it when I make a mistake.
2. I always try to practice what I preach.
3. I never resent being asked to return a favor
4. I have never been irked when people expressed ideas very different from my own (removed)
5. I have never deliberately said something that hurt someone's feelings. (removed)
6. I sometimes try to get even rather than forgive and forget. (R) (removed)

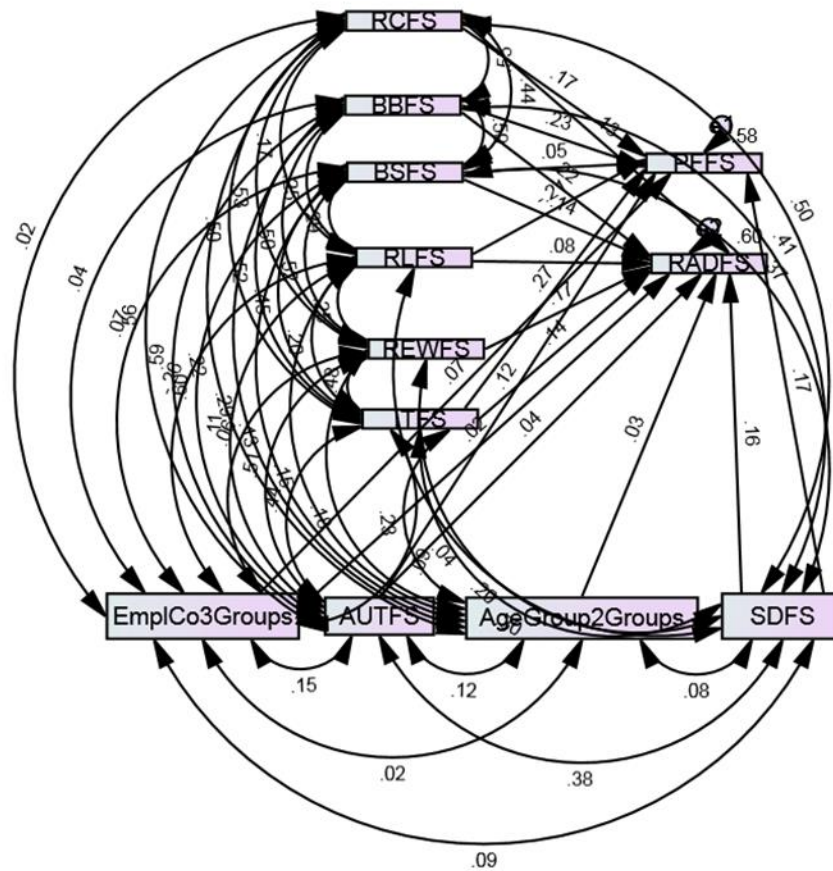
Appendix E: Additional Survey Questions

- The age of my GROUP (since generally agreed upon founding or start):
- The age of the TOTAL COMPANY (since founding):
- Our GROUP departs from the customary ways of doing business of the established TOTAL COMPANY to pursue opportunities. (LIKERT)
- The compensation/reward model is basically the same for me/my Group as it is for employees working in the more established TOTAL COMPANY.
- I am currently employed with the company/organization of the group I referred to in this survey.
- I am/will be considered for job opportunities within the TOTAL COMPANY in the future.
- The organizational form that best describes my GROUP'S type of initiative. (Check all that apply e.g., project team, business unit)

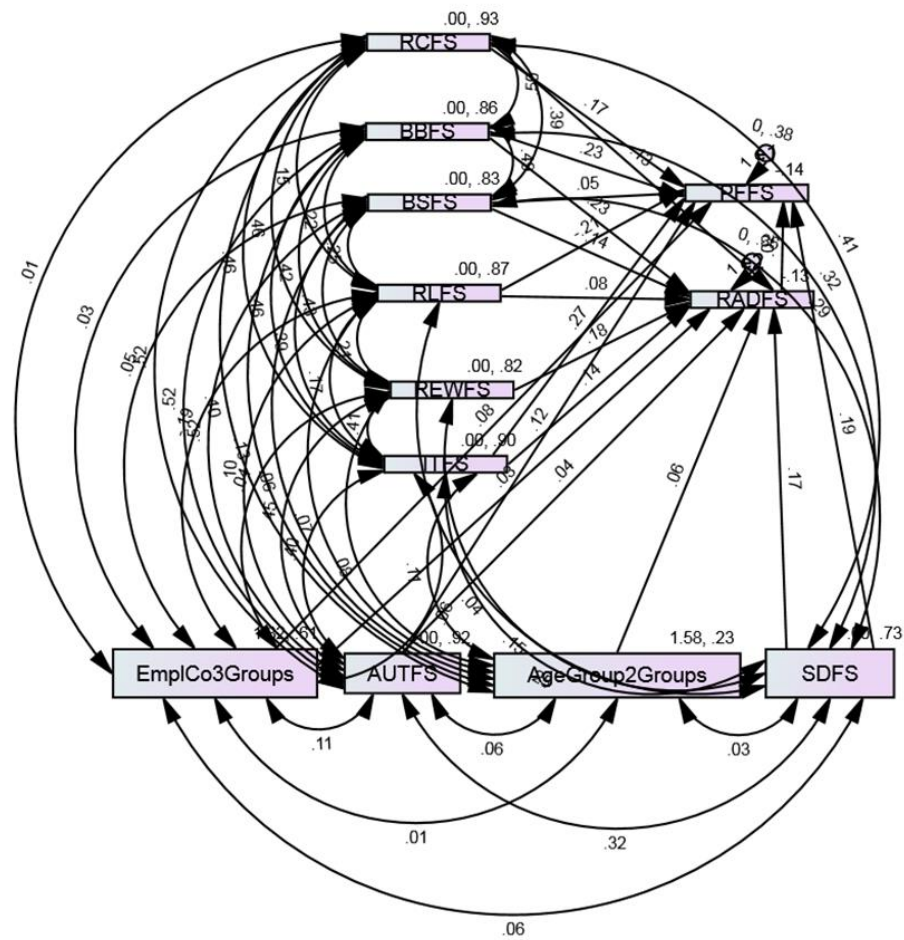
Appendix F: Incentives

	Not at all Important (1)	Extremely Important (5)	Yes (1)	No (2)
Equity in group's venture/initiative.	☐	☐	☐	☐
Access to talented people inside the Total Company	☐	☐	☐	☐
Total Company reputation and prestige	☐	☐	☐	☐
Access to financial resources of Total Company	☐	☐	☐	☐
Potential job opportunities in Total Company/other units beyond the initiative.	☐	☐	☐	☐
Protection from downside risk in the event initiative would change (e.g. fails, is sold, is absorbed)	☐	☐	☐	☐
Schedule Flexibility (4)	☐	☐	☐	☐
Location Flexibility (e.g., other places; work from home)	☐	☐	☐	☐
Incentives based on group's long-term success.	☐	☐	☐	☐
Compensation/incentives customized to group's unique circumstances.	☐	☐	☐	☐
Career Progression Support	☐	☐	☐	☐

Appendix G: Final Model 2 - Aggregated SEM



Appendix H: Supplemental Analysis – Model 3 Mediation



Appendix I: Principle Component Analysis of 11 Incentives

Total Variance Explained						
Component	Initial Eigenvalues			Extraction Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	4.896	44.505	44.505	4.896	44.505	44.505
2	1.091	9.914	54.419	1.091	9.914	54.419
3	.880	7.998	62.418			
4	.845	7.686	70.104			
5	.715	6.502	76.606			
6	.543	4.940	81.546			
7	.521	4.739	86.285			
8	.482	4.385	90.670			
9	.406	3.695	94.365			
10	.352	3.197	97.561			
11	.268	2.439	100.000			
Extraction Method: Principal Component Analysis.						

Appendix J: Summary of Open-Ended Comments

Monetary/Performance based incentives
Monetary rewards
Financial rewards
Equity
Bonus for completion of products
Startup group has incentives based on new business
Bonus based on 'kibs' completed
Quarterly bonus (15%)
Individual and company perf based incentives
Financial incentives in LT success would help
Bi-yearly bonus
Really important. Inspired more innovative work
Based on group and company profitability (2)
Group performance and co profitability (2)
Long term and short term (2)
Standard (Non-standard incentives not offered)
Rewards are standardized
No special incentives (but will change next year)
No compensation/incentives were offered
I get no rewards/incentives for my work
None for our group
(group) Rewards are standardized...
Only scheduled merit increases... Feeling of possessing all downside risk without upside...creating turnover
More equal distribution of reward and recognition for team members bearing workload
Not available to all and given to some not deserving
Rewards generally aren't worth investment...of time required for projects.
Work-Life Balance (i.e., flexible hours/schedule)
Quality of life
Flexibility with family
Work from home
More paid time off
Flexible work schedule
Time-off
Best work-life balance in this company
Work from home would increase focus
Flexible work schedule – off every other Friday
Paid time off and vacation
...good except for vacation
Non-monetary recognition/rewards
Recognition for results produced

Rewards split between groups and customers
Incentive of knowing they can take risk, but... 'keep job'.
Caregiver incentive program
Service credit
Medals
Bi-yearly rewards
Gifts
Commitment to our project and fellow team members
Miscellaneous
Opportunities for advancement
A lot of risk...incentive of still have job opportunity, if not successful.
Visibility into end-user products
Innovative company
Talented group of people
Career development program
Freedom
Learn from failure

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